



Mineralogy, mineral chemistry, and genesis of Cu-Ni-As-rich ores at Lisheen, Ireland

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Received: 24 January 2024 / Accepted: 15 July 2024 / Published online: 15 August 2024
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Abstract

The Irish Orefield is characterised by the presence of both Zn-Pb- and Cu-Ni-As-rich deposits, prospects, and orebodies in similar structural and stratigraphic positions. However, the genetic relationships between these mineralisation types are still debated. In this article, we present new mineralogical, paragenetic, and mineral-chemical observations from the Cu-Ni-As-rich ores at the classic Lisheen deposit, County Tipperary. These observations indicate the intimate association and cogenetic nature of these ores with the more abundant Zn-Pb-rich mineralisation. Specifically, both mineralisation types appear to have formed at the same time, under similar physicochemical conditions, and from the same ore fluids. In addition, both types of mineralisation contain elevated Ge contents. The cogenetic nature of the two mineralisation types, the relative absence of Cu-Ni-As-rich ores from most of the larger Irish-type Zn-Pb deposits compared to expectations derived from probable ore fluid compositions, and finally, the known geological characteristics of larger Cu-Ni-As-rich ore bodies, like Gortdrum, indicate that significant Cu-Ni-As-rich mineralisation could be present at lower stratigraphic levels across the Irish Orefield. Areas with extensive known Zn-Pb mineralisation are expected to be particularly prospective for such ores, which may occur at stratigraphic levels as deep as the Old Red Sandstone. This may have additional implications beyond Ireland, and could point to the potential for undiscovered Cu-rich ores in low-temperature carbonate-hosted Zn-Pb districts elsewhere.

Keywords Irish-type deposits · Mississippi Valley-type deposits · MVT · Copper · Germanium

Introduction

Low-temperature carbonate-hosted Zn-Pb deposits account for ~8% of global Zn, and ~12% of global Pb production (ILZSG 2017; Frenzel et al. 2017). They are also important sources of various by-products such as Ag, Co, Cd, Ga, and

Ge (Leach et al. 1995, 2005; Paradis et al. 2007; Frenzel et al. 2014, 2016a).

Most low-temperature carbonate-hosted Zn-Pb deposits have a relatively simple mineralogy, with sphalerite and galena as the major ore minerals, and various carbonates ± quartz, fluorite, pyrite, and/or marcasite as the major gangue minerals (Leach et al. 2005). However, some deposits such as those of the Viburnum Trend, USA (Cavender et al. 2016), the Irish Midlands (Hitzman and Beaty 2003; Ashton et al. 2023), Kipushi (Höll et al. 2007), and Tsumeb, Namibia (Lombaard et al. 1986), may also contain orebodies rich in Cu ± Ni, Co, and/or As, with variable mineralogy and geochemical composition. These orebodies have been interpreted by different authors as either reflecting the presence of leachable Cu-(Ni, Co, As)-rich rocks in the source regions for the deposits (e.g., mafic and ultramafic rocks, Horrall et al. 1993), or alternatively as indications for the involvement of high-temperature (> 250 °C) fluids in ore formation (e.g., Fusciardi et al. 2003; Wilkinson et al. 2005a; Torremans et al. 2018; Wilkinson 2023).

Editorial handling: D. Banks

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In addition to their genetic significance, the Cu-rich ores in these systems may also contain critical by-products. At the Kipushi and Tsumeb deposits, for instance, the Cu-rich ores are enriched in Ge and Ga, which occur mostly in the forms of germanite, renierite, gallite, and/or traces in other sulfide minerals (Lombaard et al. 1986; Höll et al. 2007). As a result, ores from these deposits have historically been an important source of these critical elements (Höll et al. 2007). In the Irish ore-field, Cu-rich ores are often enriched in Ag (Ashton et al. 2023).

The Irish Orefield is particularly interesting since it contains deposits covering the entire range from Cu(-Ag)-rich to Zn-Pb-rich compositions (Ashton et al. 2023). For instance, the Gortdrum deposit (Steed 1986; Cordeiro et al. 2023), as well as the Tullacondra (Silva et al. 2022) and Aherlow (Romer 1986) prospects, contain mostly Cu(-Ag)-dominant ores with negligible Zn and Pb. The Tynagh deposit and Kilbricken prospect contain mixed Cu-Zn-Pb ores (Derry et al. 1965; Morrissey et al. 1971; Colthurst and Reed 2019), and the Lisheen and Silvermines deposits are Zn-Pb dominated but contain minor Cu-rich ores in their cores (Andrew 1986; Hitzman et al. 2002; Fusciardi et al. 2003; Torremans et al. 2018). Finally, the giant Navan deposit, except for some traces in the Tara Deep orebody (Yesares et al. 2019), contains virtually no Cu-rich ores but is exclusively a Zn-Pb deposit (Ashton et al. 2023).

The metallogenesis of Zn-Pb-rich ores in the Irish Orefield is well documented and understood (Wilkinson and Hitzman 2015; Ashton et al. 2023; Wilkinson 2023). However, relatively little work has been done to constrain the geological significance of the Cu-rich deposits, prospects, and ores across the orefield. Three recent studies have re-examined the Cu-rich Gortdrum deposit (Cordeiro et al. 2023; Dunlevy et al. 2023) and Tullacondra prospect (Silva et al. 2022), comparing them to the classic Zn-Pb-rich deposits. Before these studies, the last descriptions of such mineralisations were published in the 1980s (Romer 1986; Steed 1986).

Since neither Gortdrum nor Tullacondra contain any Zn-Pb ores (Silva et al. 2022; Cordeiro et al. 2023), comparisons of these Cu-Ag deposits with the classic Irish-type Zn-Pb deposits must rely on published information on Cu-rich ores in deposits like Lisheen. However, only sparse and sometimes contradictory information exists for such materials in the published literature, and no detailed documentation is available for their mineralogy, paragenetic relationships, and formation conditions.

This paper aims to fill this gap. Specifically, we present the first detailed mineralogical and paragenetic description of the Cu-Ni-As-rich ores at the Lisheen Zn-Pb deposit, County Tipperary, including data on the minor and trace-element contents of the major sulfide minerals. We use this

dataset to assess the physicochemical conditions under which these Cu-Ni-As-rich ores formed, as well as their genetic relationship to the Zn-Pb-rich ores. Finally, we discuss the wider implications of our observations for the genesis and exploration potential of Cu-rich deposits in the Irish Orefield, as well as the potential of Irish-type deposits as sources of by-product Ge and Ga. Some consideration is also given to the more general implications of our findings for low-temperature carbonate-hosted deposits in other districts around the world.

The Lisheen deposit

Prior to mining, Lisheen was the second largest Zn-Pb deposit in the Irish Orefield, boasting total pre-mining resources of 23 Mt @ 13 wt% Zn, 2.3 wt% Pb and 26 g/t Ag (Güven et al. 2023). It is located in the Rathdowney trend, a distinct structural feature in the Midlands carbonate belt that also hosts the Galmoy deposit and Rapla prospect (Shearley et al. 1996; Hitzman et al. 2002; Riegler and McClenaghan 2017; Güven et al. 2023). Mining began in 1999 and ceased in 2015 with the exhaustion of the resource.

The nearly flat-lying Devonian to Carboniferous rocks in the mine area comprise five major stratigraphic units: The Old Red Sandstone, basal limestone and mudstone, Ballysteen Formation, Waulsortian Limestone, and Crosspatrick Formation (Hitzman et al. 2002; Fig. 1). This sequence is underlain by older metasedimentary basement (Hitzman et al. 2002). Due to a major pre-sulfide dolomitisation event, much of the Waulsortian Limestone in the mine area is completely dolomitised (Hitzman et al. 2002). Dolomitisation also affected the top of the Ballysteen Formation in some places (Hitzman et al. 2002).

Structurally, the area is dominated by early E- to ENE-trending normal faults, forming the Lisheen Fault System (LFS, Fig. 1; Hitzman et al. 2002; Fusciardi et al. 2003). These faults generally dip to the north at 45 to >60° and show down-dip displacements of up to 220 m (Hitzman et al. 2002; Fusciardi et al. 2003). The normal faults are intersected by a later system of minor ENE-trending folds and thrusts, as well as a last generation of sub-vertical NW trending strike-slip and normal faults.

More detailed documentation of the fault system at Lisheen, as well as an updated structural interpretation, is provided by Torremans et al. (2018) and Kyne et al. (2019), showing that the mineralisation is strongly controlled by the location of breached relay ramps between the major normal fault segments of the LFS (Torremans et al. 2018; Kyne et al. 2019). From the locations of these breached relay ramps, the irregularly shaped stratiform orebodies extend northwards along the base of the Waulsortian Limestone (Torremans et

A)

Map showing the Main Zone, Derryville Zone, and Bog Zone. Key features include the Killoran Fault Zone, Derryville Fault Zone, and Lisheen Fault System (LSF). Sample locations are marked with red squares and red circles. The map includes a scale bar (500 m) and a north arrow.

B)

Cross-section A-B showing the relationship between the Waulsortian-hosted ore body and the Oolite-hosted ore body. The section illustrates the structural units and faults, including the Waulsortian Limestone, Crosspatrick Formation, and various fault types (Major normal fault, Minor normal fault, Late strike-slip fault).

Legend for Panel A:

- Waulsortian-hosted ore body
- Oolite-hosted ore body
- Waulsortian Limestone
- Crosspatrick Formation
- Major normal fault
- Minor normal fault
- Late strike-slip fault
- Sample locations

Legend for Panel B:

- Crosspatrick Formation
- Upper Wavy Laminated Unit
- Biomicrites-limestone/dolostone
- Upper Calcarenite
- Lisduff Oolite
- Lower Calcarenite
- Black Matrix Breccia
- Massive Sulphide
- Fault
- Waulsortian-hosted ore body
- Oolite-hosted ore body

The ore lenses show a distinct metal zonation, with higher Cu and Ni concentrations close to the LFS, high As and Pb concentrations with a somewhat broader footprint, and Zn and Fe values relatively evenly distributed throughout the deposit (Torremans et al. 2018; Fig. 2, and ESM

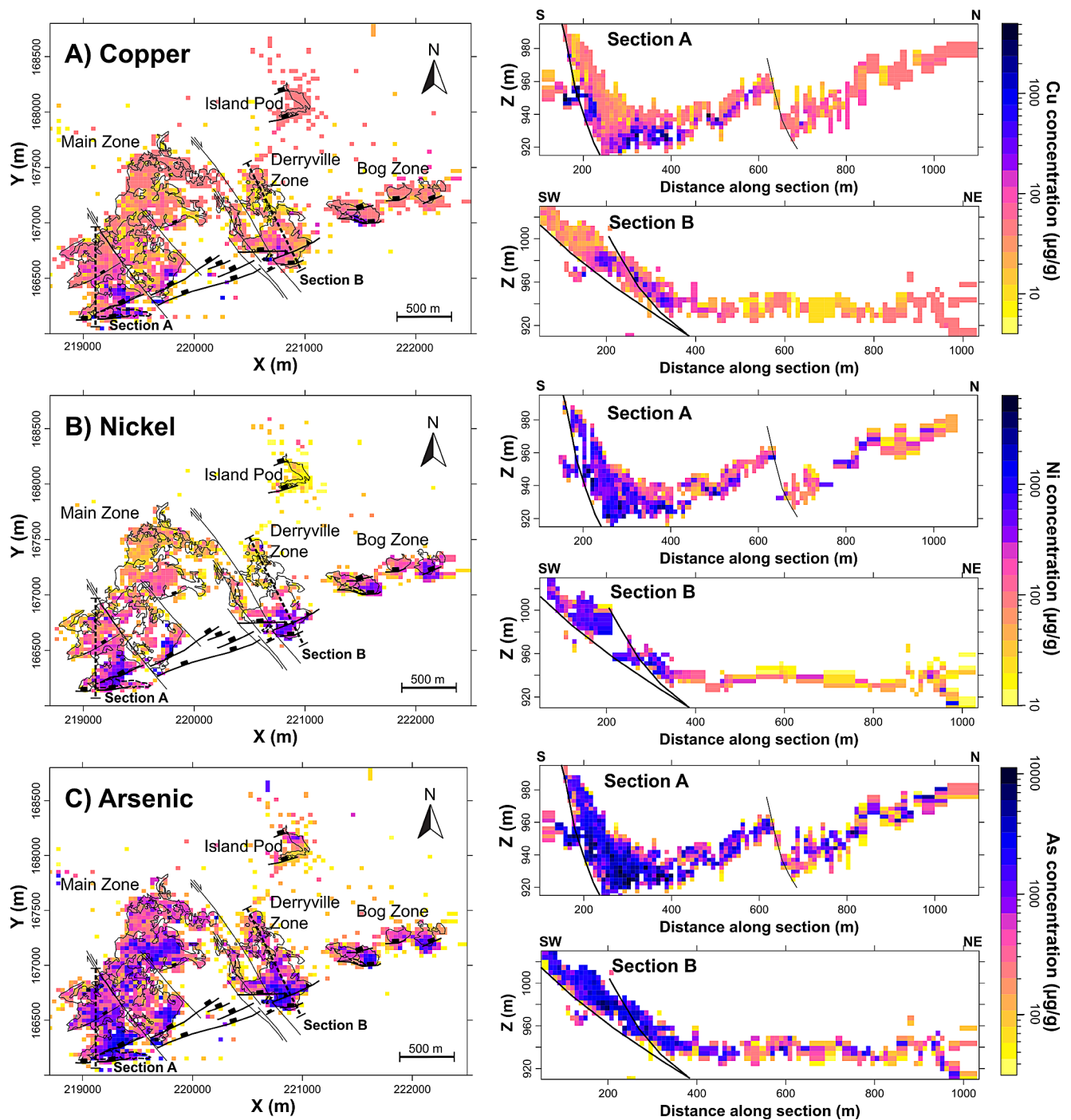


Fig. 2 Chemical zonation of the Lisheen deposit with respect to **A)** Cu, **B)** Ni and **C)** As. Surface projections (left) were compiled by taking the geometric means of drill core analyses containing more than 1 wt% total sulfide on an xy-grid with a 40×40 m bin size. This was done to provide indications of the spatial distributions of typical ore compositions rather than tonnages (as in Torremans et al. 2018). Individual samples were not weighted by length or density. Cross sections (right)

show similar geometric mean values from drill core samples within a maximum vertical distance of 20 m from either side of the section plane on a similar-sized grid. The raw data for these plots was taken from the Lisheen Mine Data Release (Irish Government 2019). Corresponding maps for Fe, Pb and Zn are shown in Fig. A1 in the electronic supplementary material (ESM Appendix A)

Fig. A1). High concentrations, particularly of Cu and Ni, are highly localised within small areas in the southern parts of the Main Zone and Derryville Zone (cf. ESM Fig. 1A, Fig. 2). For instance, the Cu-rich orebody hosting most of the Cu present in the Main Zone consisted of only 40 kt of ore at ~7 wt% Cu (Güven et al. 2023).

Torremans et al. (2018) suggest that this metal zonation reflects the flow geometry of the deep ore-forming fluid and its progressive mixing with the colder sulfur-rich reservoir: Cu, Ni, As, and Pb precipitate first, and closer to the feeder zones, and the other metals are dispersed further. This interpretation implies deposition of the Cu-Ni-As-rich ores together with the Zn-Pb-rich ores. However, the Cu-Ni-As-rich mineralisation and its relationship to the Zn-Pb-dominant ores have never been documented in detail. In fact, different sources contradict each other with respect to its paragenetic position. Fusciardi et al. (2003) describe the Cu-Ni-As-rich mineralisation as paragenetically earlier than the main-stage Zn-Pb mineralisation, while Hitzman et al. (2002) and Wilkinson et al. (2005a) describe it as paragenetically late. However, only a single micrograph is presented across all these studies to document the paragenetic position of chalcopyrite (Hitzman et al. 2002). This shows chalcopyrite, sphalerite, galena, and dolomite cross-cutting and partially replacing earlier pyrite. Overall, the available published descriptions of the relationships between the Cu-Ni-As minerals and the main ore-forming minerals pyrite-marcasite, sphalerite, and galena, are generally very short.

An internal report commissioned by the mine (Strashimirov et al. 1999) provides some more detailed micrographic and descriptive documentation of the occurrence of different Cu-Ni-As minerals, mostly in Zn-Pb-dominated ore samples. In the samples studied by Strashimirov et al. (1999), the Cu-Ni-As minerals occur mostly at trace concentrations and as inclusions in the major sulfide minerals, with a tendency towards early occurrence in the paragenetic sequence. However, this data was never published.

Materials and methods

In the following subsections, we first provide a brief overview of the mine assay database, which we used to study the distribution of different elements within the deposit and to aid in sample selection. Next, we describe the samples and sample preparation, followed by the analytical methods used to study them. The final subsection briefly explains how sphalerite chemistry was used to infer formation temperatures and sulfur fugacities for different mineral generations.

Assay database

The drillcore database of the Lisheen mine (Irish Government 2019) contains geochemical data for over 60,000 samples from 5,520 drill-cores. For the present study, we mostly examined assay values for Cu, Ni, As, Fe, Zn, and Pb and their distribution across the deposit. However, the database also contains analyses of Ag, Cd, Co, Mg, Mn, and Tl for many samples, and Ba and Hg for some. The geochemical analyses were largely conducted at ALS Loughrea by ICP-OES following a strongly oxidising acid digestion. Detection limits for this method were generally 100 µg/g for Fe, Zn, Pb, and Mg, 50 µg/g for Cu, As, Tl, and Mn, 15 µg/g for Hg, 10 µg/g for Ni, Co, and Cd, and 5 µg/g for Ag and Ba. Before using the database, we removed artefacts from data entry (mostly comma errors) and homogenised the reporting formats for values below detection limit (BDL) and missing at random (MAR) (cf. van den Boogaart and Tolosana-Delgado 2013; Frenzel 2023).

Samples

A large suite of samples (>300) was collected from drill-cores, underground workings, and ore stockpiles at the mine site between September and November 2015. These samples were first examined macroscopically. Fifty-nine samples representing both the proximal Cu-Ni-As-rich parts of the deposit, as well as the more widespread Zn-Pb mineralisation, were then selected for further study and prepared into polished rounds and thick sections (100 µm) at the in-house facilities of the Helmholtz Institute Freiberg for Resource Technology. The locations of these samples are indicated in Fig. 1. Further details on these samples are provided in ESM Tables A1 and A2. Since many of the Cu-Ni-As-rich ore samples were collected from surface and underground stockpiles, only approximate locations can be given, as indicated in Fig. 1. These locations are inferred based on the times of mining and sampling, and the known locations of the blocks of ore mined at specific times. Free access to this freshly mined material provided us with the unique opportunity to compile a comprehensive collection of these unusual ores. Scans of some representative sample slabs are shown in the ESM (ESM Figs. A2, A3).

Petrography

Initial petrographic characterisation was done in reflected light and darkfield illumination using a Nikon Eclipse LV100 Pol microscope equipped with a Prior Proscan III motorised stage. Optical scans at a resolution of 10 µm/px were recorded for further documentation using the same instrument. To identify unknown minerals and clarify

paragenetic relationships, a subset of the samples was carbon coated and examined using an FEI Quanta 450 F scanning electron microscope equipped with a Bruker Quantax EDX detector, housed at Adelaide Microscopy at the University of Adelaide.

Laser ablation-ICP-MS

Based on mineralogy, paragenesis, and spatial position within the deposit (in relation to the Cu-Ni-As-rich mineralisation), eight samples were selected for LA-ICP-MS analysis (ESM Table A3). Pyrite and sphalerite were analyzed in all samples, using an ESI NWR213 solid state laser coupled to an Agilent 7900 ICP-MS (Adelaide Microscopy). In addition, analyses were conducted for galena, chalcopyrite, bornite, and tennantite where these were present.

A minimum of five spot analyses were performed on each mineral generation in each sample. The following isotopes were monitored during each measurement (30 s background, 50 s ablation): ^{55}Mn , ^{57}Fe , ^{59}Co , ^{60}Ni , ^{63}Cu , ^{66}Zn , ^{69}Ga , ^{72}Ge , ^{73}Ge , ^{75}As , ^{77}Se , ^{107}Ag , ^{111}Cd , ^{113}In , ^{115}In , ^{118}Sn , ^{121}Sb , ^{125}Te , ^{202}Hg , ^{205}Tl , ^{208}Pb , and ^{209}Bi . Ablation spot diameters ranged from 20 to 50 μm to accommodate textural variability within and between samples, avoiding inclusions.

To convert the measured signals to concentrations, ^{57}Fe (pyrite, chalcopyrite, bornite), ^{63}Cu (tennantite), ^{66}Zn (sphalerite), and ^{208}Pb (galena) were used as internal standards, with MASS-1 (Wilson et al. 2002) as the external standard material. Stoichiometric Fe and Pb contents were assumed for pyrite, chalcopyrite, bornite, and galena. For sphalerite, a Zn concentration of 66 wt% was assumed, given the universally low Fe contents (mostly < 1 wt%; cf. Wilkinson et al. 2005a). For tennantite, a Cu concentration of 47.5 wt% was assumed for initial data reduction. Initial estimates of Fe, Zn, and Sb concentrations were then used together with As and S contents estimated from stoichiometric constraints to derive actual Cu contents in the analysed tennantite samples and use these to correct all analysis results (cf. method in Swinkels et al. 2021).

A block of three standard measurements was inserted after every 20 measurements on a sample. Accuracy was monitored by including one measurement on NIST610 glass (Jochum et al. 2011) with every standard block. Offline corrections were made for the isobaric interferences of ^{113}Cd on ^{113}In , and ^{115}Sn on ^{115}In . The magnitude of the molecular interferences of $^{56}\text{Fe}^{16}\text{O}$ and $^{57}\text{Fe}^{16}\text{O}$ on ^{72}Ge and ^{73}Ge , respectively, were monitored using the count ratio of ^{72}Ge to ^{73}Ge (cf. Frenzel et al. 2020). Inclusions were identified by examining individual ablation traces for characteristic irregularities. Integration intervals were chosen to avoid inclusions. Measurements of inclusion-rich materials where

inclusions could not be excluded were rejected. In addition, typical inclusion mineral assemblages were documented during petrographic characterisation. Data presentation and analysis followed the recommendations and guidelines outlined in Dmitrijeva et al. (2018), Godefroy-Rodriguez et al. (2020) and Frenzel (2023).

Sphalerite thermometry and fS_2 reconstruction

Sphalerite formation temperatures were estimated using the GGIMFis geothermometer (Frenzel et al. 2016b). This is based on an empirical relationship between sphalerite composition and formation temperature as recorded by fluid inclusion homogenisation temperatures, and is described by the following equation:

$$T(^{\circ}\text{C}) = -(54.4 \pm 7.3) \cdot PC1^* + (208 \pm 10) \quad (1)$$

with

$$PC1^* = \ln \left(\frac{c_{Ga}^{0.22} \cdot c_{Ge}^{0.22}}{c_{Fe}^{0.37} \cdot c_{Mn}^{0.20} \cdot c_{In}^{0.10}} \right) \quad (2)$$

Where $\ln(x)$ denotes the natural logarithm (base e), and c_i is the concentration of trace element i in sphalerite, given in units of $\mu\text{g/g}$ for Ga, Ge, Mn, and In, and in wt% for Fe. Uncertainties on the absolute temperatures estimated from (1) are generally on the order of $\pm 50^{\circ}\text{C}$ across the calibration range from 100 to 400°C . The GGIMFis geothermometer is expected to work well when estimating the average formation temperatures of distinct mineralisation events within individual deposits (cf. Frenzel et al. 2016b; Bauer et al. 2019) but may also be able to capture smaller-scale variations within individual samples (cf. Frenzel et al. 2022).

In addition to formation temperatures, we also estimated the sulphur fugacities, fS_2 , that prevailed during sphalerite formation, using an updated model presented in Frenzel et al. (2022) based on the original work of Barton and Toulmin (1966), as well as newer experimental data from Scott and Barnes (1971), Scott and Kissin (1973), and Lusk and Calder (2004). For this, the measured Fe-contents in sphalerite are first recalculated to mol% FeS. These concentrations can then be used to estimate the FeS-activity in the sphalerite, $[FeS(sp)]$, from

$$\log_{10} [FeS(sp)] = (0.80 \pm 0.02) \cdot \log_{10} \left(53.1 \cdot \frac{FeS}{FeS_{max}} \right) - (1.38 \pm 0.03) \quad (3)$$

Where FeS is the Fe-content in sphalerite in mol% FeS, and FeS_{max} is the temperature- and pressure-dependent maximum solubility of FeS in sphalerite when in contact with metallic Fe and troilite, described by:

$$FeS_{max} = 44.09 + 0.0125 \cdot T - \frac{3600}{T} \cdot p \quad (4)$$

With T – temperature in Kelvin, and p – pressure in kbar (Barton and Toulmin 1966; Balabin and Urusov 1995). A formation pressure of 0.07 ± 0.05 kbar (assuming a formation depth between 200 and 1,200 m below sea-level, cf. Hnatyshin et al. 2015, 2020) and formation temperatures estimated from the GGIMFis geothermometer were used in Eq. (4). The resultant FeS -activities were then combined with GGIMFis temperatures to calculate fS_2 values, including uncertainties, using the following model:

$$\log_{10} fS_2 = -2\log_{10} [FeS (sp)] - \frac{(1.50 \pm 0.02) \times 10^4}{T} + (14.5 \pm 0.3) \quad (5)$$

This estimation procedure for fS_2 relies on the assumption that Fe-contents in sphalerite are buffered by pyrite. Pyrite is present in virtually all our samples and is generally an abundant and widely distributed mineral at Lisheen. It occurs throughout the complete paragenetic sequence. This means that pyrite-buffering is a reasonable assumption.

A further note concerns Cu contents in sphalerite: the model is generally only considered to be valid when Cu contents are below ~0.6 wt% in the sphalerite. Otherwise, a correction term, dS_2^{Cu} , must be added to the value estimated from Eq. (3) to (5) such that:

$$df S_2^{Cu} = (3.4 \pm 1.4) \cdot \log_{10} CuS + (1.1 \pm 0.4) \quad (6)$$

Where CuS is the concentration of Cu in sphalerite, expressed in mol% CuS (Frenzel et al. 2022).

Results

The following subsections summarise the main results of this study, starting with an overview of the bulk geochemical compositions of the Cu-Ni-As- and Zn-Pb-rich ores, and proceeding with a description of the mineralogy and paragenesis of the samples. Finally, we present the corresponding results of the LA-ICP-MS analyses of the major sulfide minerals.

Geochemical composition of Cu-Ni-As- vs. Zn-Pb-rich ores

Table 1 summarises the bulk geochemical compositions of the Cu-Ni-As-rich and Zn-Pb-rich ore samples from the Lisheen assay database. Only complete observations were considered for this comparison, where all the reported elements had been analysed. This corresponds to 40,995 samples or about two thirds of the entire drill-core database. The

Table 1 compositions of Cu-rich vs. Zn-Pb-rich ores (from mine database)

Ore type	<i>n</i>	Ag (μg/g)	As (μg/g)	Cd (μg/g)	Co (μg/g)	Cu (wt%)	Fe (wt%)	Hg (μg/g)	Mg (wt%)	Mn (μg/g)	Ni (μg/g)	Pb (wt%)	Tl (μg/g)	Zn (wt%)
Cu	1,249	1st Qrtl.	40	2,580	120	143	0.18	4.53	<15	0.08	150	1.08	<50	2.94
		Median	88	5,364	280	319	0.36	12.7	<15	0.75	280	3.04	150	7.22
		Mean	144	12,400	817	580	1.43	14.6	<15	2.49	390	7.03	209	11.7
		3rd Qrtl.	175	10,860	700	662	1.09	22.8	<15	3.90	530	9.14	286	18.3
Zn	17,829	1st Qrtl.	<5	1,240	79	105	<0.005	6.95	<15	0.04	103	0.64	115	8.18
		Median	8.4	2,212	174	190	<0.005	18.2	<15	0.35	192	1.43	276	15.2
		Mean	31	2,580	309	220	0.010	18.5	<15	2.54	265	2.75	308	18.5
		3rd Qrtl.	38	3,212	370	286	0.007	29.1	<15	4.71	362	3.15	446	26.6

Note Only the subset of data without MAR values was considered for the summary presented in this table ($N=40,995$). Copper-rich ores were taken as those with a minimum of 0.1 wt% Cu, while Zn-rich ores were defined as those with a maximum of 0.1 wt% Cu, and a minimum of 5 wt% combined Zn and Pb. The reported mean values correspond to the arithmetic means for the elements

Cu-Ni-As-rich ore samples were then defined as those with > 0.1 wt% Cu, while Zn-Pb-rich ore samples were defined as those with < 0.1 wt% Cu, and > 5 wt% combined Zn and Pb. The summary statistics correspond to these two groups, with overall sample numbers reported under column *n* in Table 1. Values below detection limit were replaced with the respective detection limits for the evaluation of the reported mean values (arithmetic means).

It is apparent from this summary that the Cu-Ni-As-rich ores contain substantially higher concentrations of Ag, As, Cd, Co, Mn, Ni, and Pb on average than the Zn-Pb-rich ores, which in turn are relatively enriched in Fe, Tl, and Zn. Contents of Mg are comparable. Mercury contents are below the detection limit of 15 µg/g across virtually all samples, and therefore, no clear trend can be discerned.

To provide an idea of the variability between samples, Table 2 shows the compositions of the individual drill-core intervals selected for the preparation of polished samples. Readers will note that substantial variations occur within the two groups of samples and that there is considerable overlap in the ranges of most of the reported elements. However, the relative differences in median values of all elements between the Cu-Ni-As-rich and Zn-Pb-rich samples are similar to those seen for the entire deposit (Table 1). However, the samples selected for this study are generally richer in the sulfide-hosted elements (Cu, Zn, Pb, Fe, Ag etc.).

Ore mineralogy, textures, and paragenesis

Figure 3 shows a set of reflected-light scans of polished sections, which are broadly representative of the range in mineralogical compositions and ore textures observed in this study. ESM Tables A1 and A2 summarise the modal mineralogical compositions of all the 59 polished samples studied in detail.

While the mineralogy of the Cu-Ni-As-rich samples shows considerable variability (Fig. 3A–I), the Zn-Pb-rich samples (Fig. 3J–H) are typical of other low-temperature carbonate-hosted Zn-Pb deposits. The dominant sulfide minerals in the Zn-Pb-rich ores are pyrite, marcasite, sphalerite, and galena (Fig. 3J–H). Other sulfides, such as tennantite and chalcopyrite only occur rarely and in trace amounts. The main gangue minerals in the Zn-Pb-rich samples are dolomite and calcite, some quartz and other silicate minerals, as well as rare barite. In exceptional cases, other minerals are also present, such as vaesite in sample Li-HS-12.

The Cu-rich samples generally contain the same ore and gangue minerals as the Zn-Pb-rich samples. However, a complex suite of Cu, Ni, and As-bearing phases is also present. The most common Cu mineral is tennantite-(Zn) (Fig. 3A, D–G; ESM Tables A1 and A2). However, chalcopyrite dominates in some samples (Fig. 3B, C), and bornite

may be present as an additional copper mineral (Fig. 3A, B). The Ni-As minerals occur as a variable association of mostly gersdorffite and (para)rammelsbergite, with sporadic niccolite, and millerite (Fig. 3G–I). Vaesite and arsenopyrite are rare, occurring only as minor or trace phases in some of the investigated samples.

The samples illustrated in Fig. 3 demonstrate that there is a continuous spectrum in sulfide mineralogy from samples containing mostly the Cu-Ni-As minerals, via mixed samples, to the more widespread Zn-Pb-rich ores dominated by pyrite, marcasite, sphalerite, and galena. This mirrors the relatively smooth gradation from Cu-Ni-As-rich to Zn-Pb-rich ores seen in the large-scale spatial distributions of the corresponding elements (cf. Figure 2, ESM Fig. A1).

Ore textures and the intergrowth relationships between different mineral generations are complex and dominated by space-filling, replacement, and brecciation textures. Some samples have also been deformed plastically. Figures 4, 5, and 6 present a variety of optical and back-scattered electron (BSE) micrographs to illustrate this. In general, the same sulfide mineral textures occur in both the Cu-Ni-As-rich and Zn-Pb-rich samples.

Space-filling textures range from euhedral crystals via dendritic and skeletal growths to fine-grained colloform layers. Colloform and dendritic textures are common for the Cu-Ni-As minerals, often in intimate association with each other and the other sulfides (Fig. 4B–D, F and I–L). Space-filling pyrite(-marcasite) textures range from coarsely crystalline columnar growths (Fig. 5A) to finer-grained colloform-banded (Fig. 6F) and sometimes skeletal aggregates. Space-filling sphalerite is generally colloform (Fig. 5B–E, G and H) to arboriform (Fig. 4D), but sometimes occurs as coarsely crystalline (e.g., Sp I in sample Li-HS-86; ESM Fig. A5A, B) or fine granular masses. Galena occurs as subhedral to euhedral crystals (Fig. 5K), and sometimes skeletal or dendritic growths (Figs. 4G and 5H). In many samples, space fills occur, which have the appearance of internally deposited sediments, mostly composed of layers of fine-grained sphalerite, carbonates, and/or pyrite (e.g., Fig. 5K, L).

Replacement textures are most common where sphalerite, pyrite, and/or other sulfides replace host-rock fragments or fossils, such as crinoid fragments, (Figs. 3N and 5F and J). Chalcopyrite, sometimes with galena, is often seen to replace earlier columnar to colloform pyrite-marcasite (Fig. 4C and E), and tennantite sometimes replaces earlier sphalerite, galena, and pyrite (Fig. 4G) or barite (Fig. 4A). Otherwise, extensive replacement of one sulfide by another was not observed in the present sample suite.

Pre- to syn-mineral *fracturing* or *brecciation* is seen in the majority of samples, with clasts of fractured host-rocks and/or sulfides overgrown and cemented by later sulfide

Table 2 Geochemical assays of sampled drill-core intervals (from mine database)

Orebody ¹	Ore type	Hole No. ²	Interval start (m)	Length (m)	Ag (μg/g)	As (μg/g)	Ba (μg/g)	Cd (μg/g)	Co (μg/g)	Cu (wt%)	Fe (wt%)	Hg (μg/g)	Mg (wt%)	Mn (μg/g)	Ni (μg/g)	Pb (wt%)	Tl (μg/g)	Zn (wt%)
MZ (S)	Cu	LK-0124	230.7	0.5	204	20,200	44,190	735	n.d.	3.71	25.0	0.4	<0.01	n.d.	n.d.	6.05	n.d.	12.25
MZ (S)	Cu	LK-0179	175.25	0.46	210	60,000	50	778	n.d.	11.53	11.6	0.4	<0.01	n.d.	n.d.	9.88	n.d.	21.7
MZ (S)	Cu	LK-0179	176.75	0.4	44	6,400	90	198	n.d.	0.81	28.5	0.3	<0.01	n.d.	n.d.	2.06	n.d.	8.5
MZ (S)	Cu	LK-0311	217	0.5	22	2,680	n.d.	78	n.d.	0.71	4.0	n.d.	<0.01	n.d.	n.d.	2.64	n.d.	1.66
MZ (Oo)	Cu	LK-0985	182.4	1	34	243,109	n.d.	170	3,319	0.125	8.77	<5	0.31	180	175,740	1.28	103	6.36
MZ (Oo)	Cu	LK-1004	167.6	1.1	141	16,730	n.d.	196	611	1.88	19.16	<5	1.09	300	6,358	5.63	146	9.04
MZ (Oo)	Cu	LK-1004	170.7	1	58	6,280	n.d.	243	295	0.88	5.84	<5	1.51	230	2,697	1.58	88	8.27
MZ (S)	Cu	LK-1114	191.5	0.9	2,128	150,950	n.d.	472	2,287	10.2	12.41	<15	0.22	120	56,726	0.71	170	4.03
MZ (S)	Cu	LK-1114*	192.4	1	1,743	173,290	n.d.	439	2,177	10.7	9.2	<15	0.19	120	79,046	0.69	122	3.3
MZ (Oo)	Cu	LK-1145	169.4	0.9	132	30,280	n.d.	123	818	1.56	34.25	<15	0.11	<50	11,310	1.98	321	6.39
MZ (Oo)	Cu	LK-1151	170.7	0.3	919	20,590	n.d.	6,843	228	4.44	9.83	21	0.01	110	506	9.14	182	35.81
MZ (Oo)	Cu	LK-1155	171.1	0.6	137	6,500	n.d.	880	278	1.43	3.01	<15	1.41	200	607	9.09	289	24.34
MZ (Oo)	Cu	LK-1485	178.1	1.15	162	8,990	n.d.	891	355	1.70	9.0	<15	0.08	20	3,478	4.34	218	35.32
MZ (Oo)	Cu	LK-1833	165.2	0.95	571	20,870	n.d.	725	527	3.96	7.16	<15	<0.01	20	4,654	21.83	341	21.15
BZ	Cu	LK-1910	160.6	0.6	161	2,480	n.d.	5,292	56	3.14	4.62	<15	0.36	220	36	23.13	<50	18.4
MZ (S)	Cu	MSC-431	7.5	0.9	13	2,550	n.d.	<10	200	0.21	7.08	<15	6.52	740	747	0.12	146	1.62
Median composition (Cu-rich intervals):																		
MZ (S)	Zn	LK-0124	234.2	0.5	95	5,100	3,000	485	n.d.	0.068	35.5	0.05	<0.01	n.d.	n.d.	4.94	n.d.	7.1
MZ (S)	Zn	LK-0457	211	0.5	531	13,000	n.d.	690	1,880	0.09	22.4	n.d.	1.05	n.d.	5,600	13.53	390	12.44
DZ (N)	Zn	LK-0644*	184.3	1	85	3,420	n.d.	520	40	0.02	32.65	n.d.	0.13	n.d.	80	4.65	290	18.75
DZ (C)	Zn	LK-0716	171	1.1	23	2,410	n.d.	58	160	0.005	30.5	n.d.	0.03	n.d.	510	0.84	210	7.72
DZ (C)	Zn	LK-0722	164.8	1	80	2,660	n.d.	256	480	0.011	28.6	n.d.	0.03	n.d.	1,700	0.98	510	21.61
DZ (C)	Zn	LK-0873	165.3	0.9	79	1,872	n.d.	215	355	0.002	29.3	<5	0.03	70	1,340	2.49	761	18.38
DZ (E)	Zn	LK-0885	135.75	1	1	2,320	n.d.	40	223	<0.001	23.5	<5	0.03	50	591	3.24	389	31.62
DZ (S)	Zn	LK-0910	136.25	1	83	2,998	n.d.	286	95	0.046	38.2	<5	0.02	180	386	6.27	210	11.16
DZ (S)	Zn	LK-0910	131.25	1	126	3,039	n.d.	124	258	0.011	30.51	<5	0.02	120	755	1.61	471	25.04
DZ (C)	Zn	LK-0915	193.9	0.5	13	1,161	n.d.	335	84	<0.001	29.12	<5	0.30	50	192	2.24	50	22.86
DZ (S)	Zn	LK-0935	200.5	0.5	65	2,332	n.d.	829	80	0.004	16.64	<5	0.02	150	183	9.42	155	33.87
DZ (N)	Zn	LK-1010	183.6	0.9	<2	230	n.d.	5	15	<0.001	5.94	<5	9.22	480	50	0.038	<25	0.537
MZ (S)	Zn	LK-1016	194.3	0.7	52	11,520	n.d.	568	289	0.072	21.94	<5	1.17	380	1,947	4.65	465	15.81
MZ (S)	Zn	LK-1102	155.2	1	<5	13,090	n.d.	56	1,642	<0.005	20.42	<15	0.03	<50	5,206	1.17	342	30.32
DZ (N)	Zn	LK-1290*	190.6	0.8	14	2,970	n.d.	212	102	<0.005	19.3	<15	4.83	190	200	0.38	345	13.67
BZ	Zn	LK-1794*	156.4	1.3	<5	4,480	n.d.	194	358	<0.005	29.07	<15	0.05	110	847	2.08	780	16.46
Median composition (Zn-Pb-rich intervals):																		
					52	2,984	n.d.	236	223	0.005	28.84	<5	0.03	135	591	2.37	345	16.14

Note n.d. – not determined

¹Ore body abbreviations: BZ – Bog Zone, DZ – Derryville Zone, MZ – Main Zone (cf. Figure 1); the classifiers in parentheses denote: C – Central, E – East, N – North, S – South, Oo – Oolite²Samples from the core intervals marked with * were analysed further by LA-ICP-MS

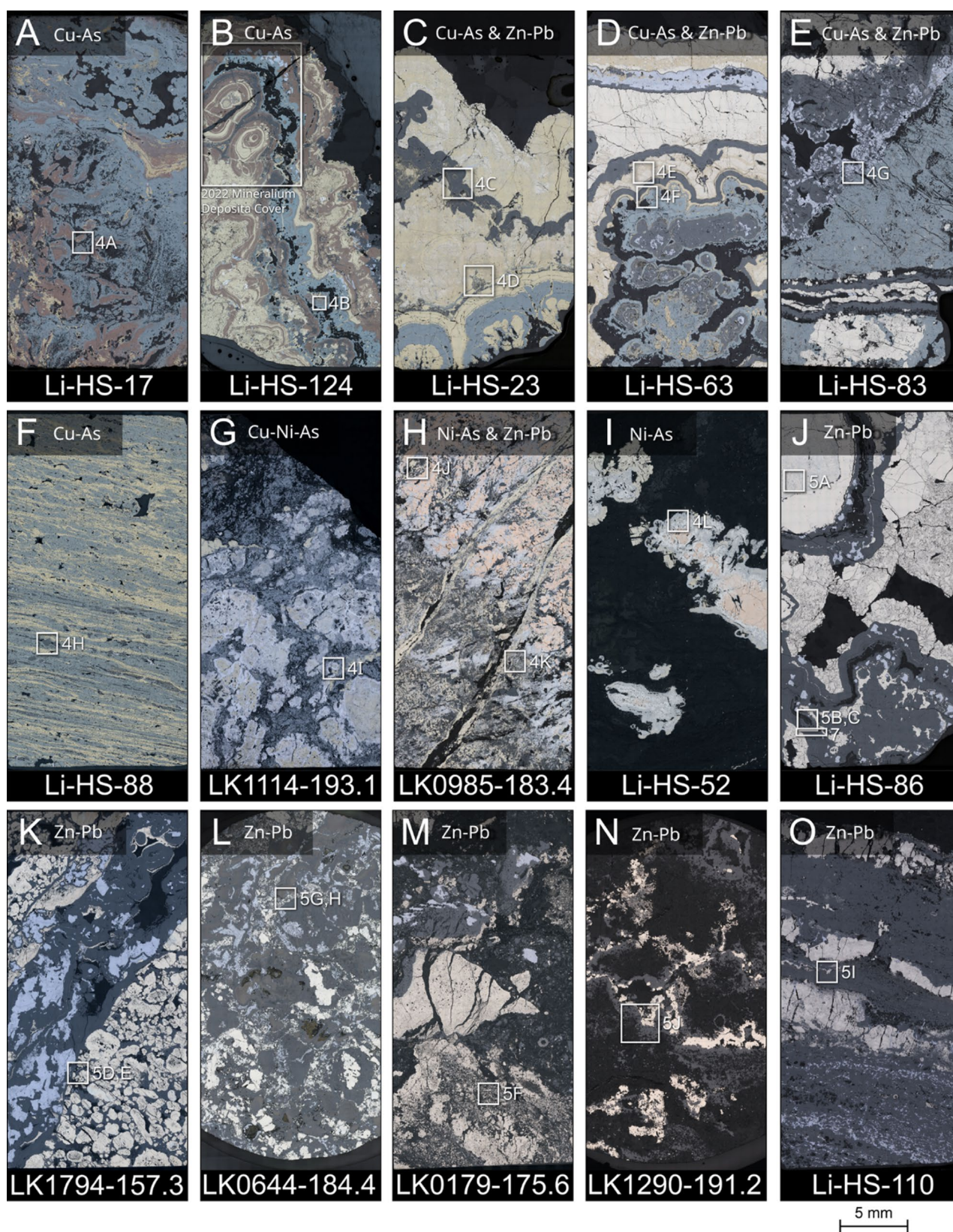


Fig. 3 Microscope scans of selected polished ore sections (plane-polarised reflected light) with locations of more detailed micrographs in Figs. 4, 5, and 6 indicated by the square frames: **A)** Cu-rich sample containing mostly tennantite, bornite, and chalcopyrite. The transparent mineral (in black) is mostly barite. **B)** Cu-rich sample from Main Zone (South / Oolite), containing abundant colloform chalcopyrite, bornite, and tennantite, with some sphalerite, pyrite, and galena. The transparent mineral is again barite. **C)** Cu-rich sample with mostly chalcopyrite and tennantite, in addition to colloform sphalerite and pyrite. The transparent minerals are early barite (bottom) and late calcite (top). **D)** Sample similar to C) from Main Zone (South / Oolite), again with early barite (bottom) and late calcite (top). **E)** Cu-rich sample from the Derryville Zone (Oolite), containing tennantite, pyrite-marcasite, sphalerite, and galena as the only sulfide minerals. The transparent gangue minerals are mostly dolomite and calcite. **F)** Plastically deformed Cu-rich sample from the Derryville Zone (Oolite) consisting mostly of chalcopyrite and tennantite. **G)** Nickel-rich sample from the Main Zone (South) containing abundant gersdorffite and (para-)rammelsbergite, in addition to tennantite, chalcopyrite, and minor sphalerite, pyrite, and galena. **H)** Nickel-rich sample from the Main Zone (Oolite) containing abundant early nickeline (**upper right**) with later sphalerite, pyrite, and galena (**bottom left**). The transparent minerals are mostly barite and carbonates. **I)** Nickel-rich sample containing mostly nickeline, gersdorffite, (para-)rammelsbergite, and minor millerite, with some barite in a carbonate matrix. **J)** Zn-Pb rich sample from the Main Zone (South / Oolite), containing mostly colloform sphalerite, pyrite, and crystalline galena. The transparent mineral is mostly calcite. **K)** Zn-Pb-rich sample from the Bog Zone, consisting mostly of colloform sphalerite, pyrite, and crystalline galena, with carbonate gangue. **L)** Highly brecciated Zn-Pb-rich sample from the Derryville Zone (Central) consisting mostly of fragments of intergrown colloform sphalerite, euhedral to dendritic galena, and pyrite. **M)** Zn-Pb-rich sample from the Main Zone (Oolite) containing abundant early iron sulfides and showing signs of syn- to post-mineral brecciation. **N)** Zn-Pb-rich sample from the Derryville Zone (North) showing sulfides as cements and clast-replacements in dolomite breccia. **O)** Plastically deformed Zn-Pb-rich sample from the Derryville Zone (South). Note that sample Li-HS-124 (panel B) was featured on the 2022 Mineralium Deposita magazine cover as indicated

generations (e.g., Fig. 3G, H and L – N, 4A, 5G, H; also ESM Fig. A4). *Plastic deformation textures*, in contrast, are rare and are restricted to samples consisting mostly of ductile minerals from within or near the major fault zones of the LFS (e.g., Figs. 3F and O, 4H and 5I).

Most samples show evidence for cyclical mineral deposition. Multiple alternating generations of the sulfide and gangue minerals are usually present, often with repetitions of several generations of a given mineral or mineral association (e.g., sphalerite, Fig. 3 and ESM Fig. A5). Generally, the different minerals appear to have precipitated in separate pulses within individual samples, with the exception of (1) sphalerite and galena, and (2) the Cu-Ni-As-minerals and barite, which often precipitated together or in very close spatiotemporal association.

The general observations on sulfide mineralogy and textures summarised above hold true for both the Waulsortian- and Lisduff-Oolite-hosted mineralisations. However, host-rock replacement textures were more commonly observed in the Lisduff-Oolite, particularly for early mineral

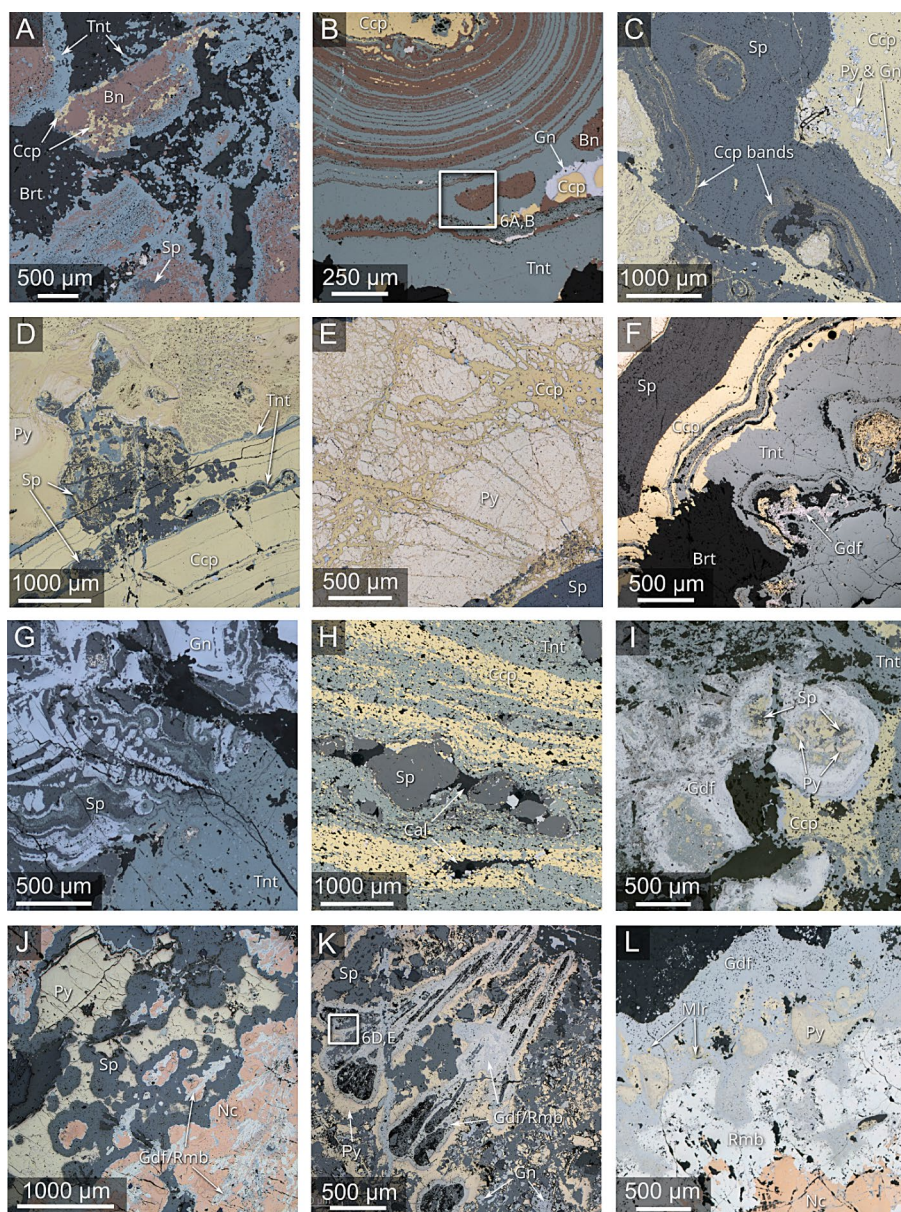
generations. Due to the different host rocks, fine-grained shaley material is also more abundant in the oolite ores. Finally, the most Ni-rich samples we examined came from the oolite-hosted orebodies.

Despite the relatively clear paragenetic sequences present in individual samples, it is practically impossible to correlate specific mineral generations across samples from different parts of the deposit. This is mostly due to (1) the absence of distinct “marker” generations, (2) the fact that the mineralogical characteristics of contemporaneous mineral generations appear to vary across the deposit (e.g., chalcopyrite-tennantite in one place may have been deposited at the same time as sphalerite or pyrite in another), and finally, (3) the incompleteness of the sequences represented within individual samples. Figure A5 illustrates some of these difficulties using two samples from the southern part of the Main Zone orebody. Because of these difficulties in direct correlation, mineral generation labels used in the subsequent parts of this manuscript are sample-specific and only denote the relative order of events within a sample, i.e., Sp I in one sample is not the same as Sp I in another.

Despite these difficulties, it is possible to use the petrographic results from the entire sample set to identify some general temporal trends in mineral deposition. An early iron-sulfide dominated phase, similar to that described by previous authors (Hitzman et al. 2002; Fusciardi et al. 2003), is also present in several of our samples, both from the Cu-Ni-As-rich and Zn-Pb-rich sample groups. During the subsequent main-stage mineralisation, the Cu-Ni-As-rich and Zn-Pb-rich mineral assemblages generally show an intimate temporal association, often with alternations between the two types of assemblages where they occur together in the same sample. Overall, however, there is a tendency for the Cu- and Ni-minerals to dominate early in the paragenetic sequence in such samples, while pyrite, sphalerite and galena dominate later (e.g., in samples Li-HS-23, Li-HS-63, and LK-0985-183.4 in Fig. 3). Exceptions to this rule occur, e.g., in sample LK-1114-193.1, where some early sphalerite and pyrite predate the Cu-Ni-As minerals. Furthermore, some Zn-Pb-rich samples contain minor late-stage chalcopyrite-galena veins, which crosscut all other mineral generations, as the only evidence of Cu-rich mineralisation. In most cases, however, the late veins mostly contain relatively coarse crystals of honeyblende sphalerite and final calcite cement, which conclude the paragenetic sequence. These general trends hold for both the Waulsortian- and the Lisduff Oolite-hosted samples.

Because the Cu and Ni minerals are essentially absent from most samples of the more distal Zn-Pb dominant mineralisation, it is difficult to determine their relative temporal position with respect to such samples. However, Cu-Ni-As-Zn-rich growth zones occurring within pyrite (e.g., Fig. 6F,

Fig. 4 Selected micrographs showing details of the Cu-Ni-As-rich samples presented in Fig. 3. **A)** Detail of sample Li-HS-17 showing some evidence of fragmentation of earlier colloform bornite-chalcopyrite-tennantite intergrowths, rimmed and cemented by tennantite and barite. **B)** Detail of colloform-layered intergrowths of bornite and tennantite in sample Li-HS-124. **C)** Colloform sphalerite with interbands rich in chalcopyrite inclusions on sample Li-HS-23. **D)** Arboriform sphalerite growth interspersed with colloform-layered chalcopyrite and tennantite (sample Li-HS-23). **E)** Fractured colloform to columnar pyrite partially replaced by chalcopyrite with minor galena in sample Li-HS-63. **F)** Colloform-layered intergrowths of tennantite, sphalerite, chalcopyrite, and gersdorffite, associated with bunches of lath-like barite crystals in sample Li-HS-63. **G)** Tennantite replacing colloform sphalerite intergrown with dendritic to colloform galena. **H)** Sphalerite ‘porphyroblasts’ with calcite-filled strain shadows in plastically deformed matrix of chalcopyrite and tennantite (sample Li-HS-88). **I)** Fragments of early pyrite and sphalerite overgrown by colloform gersdorffite, with later infill of tennantite, chalcopyrite, and barite (sample LK1114-193.1). **J)** Earlier dendritic gersdorffite / (para-)rammelsbergite overgrown by colloform nickeline, followed by colloform sphalerite and coarse pyrite (sample LK0985-183.4). **K)** Acicular to dendritic growth of gersdorffite/(para)rammelsbergite and sphalerite overgrown by later sphalerite and galena (sample LK0985-183.4). **L)** Detail of successive colloform growths of nickeline, (para-)rammelsbergite, and gersdorffite + millerite + pyrite (sample Li-HS-52). Mineral symbols after Warr (2021)



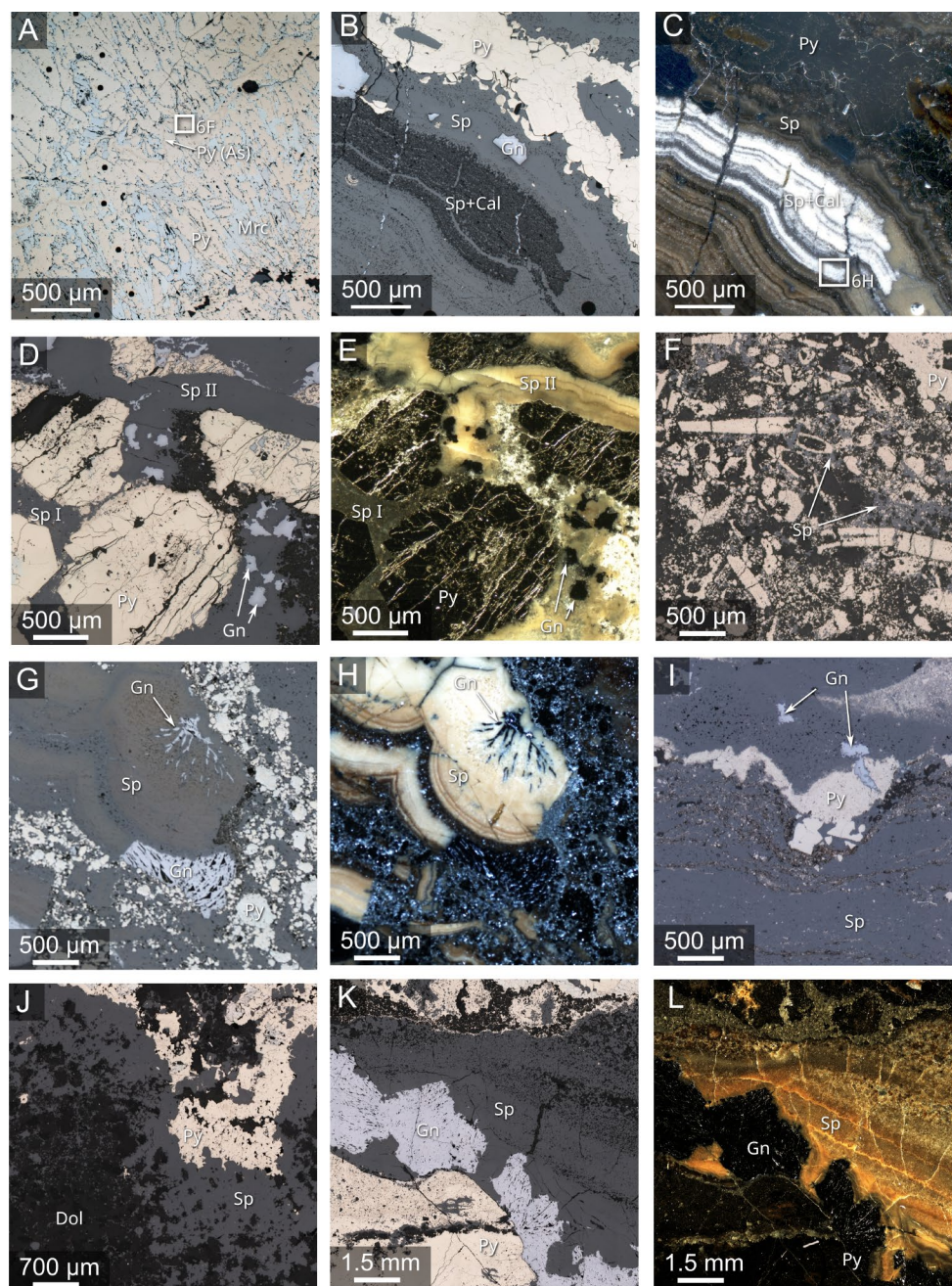
G) and sphalerite (Fig. 7) in the more distal Zn-Pb-rich samples may provide a major hint of the probable deposition of Cu-Ni-As minerals elsewhere in the system. Such growth zones tend to occur across all main-stage pyrite generations, with a slight tendency towards earlier generations. Late vein-hosted Cu minerals in distal ore samples, on the other hand, are probably the result of later remobilisation as described by Wilkinson et al. (2005a) and Torremans et al. (2018), rather than the manifestation of a late Cu input into the system.

Together, these observations indicate that the deposition of the Cu-Ni-As minerals occurred throughout the main stage of ore formation at Lisheen, with some tendency towards an early prevalence of Cu-Ni-As mineral deposition.

Micron- to sub-micron scale mineral textures

The micron- to sub-micron-scale textures of the different ore minerals are important features for the subsequent interpretation of the LA-ICP-MS data. This particularly concerns the presence of micro- to nano-inclusions of other minerals. In general, small-scale examination of the different sulfide minerals demonstrated the presence of abundant micro- to nano-scale porosity and mineral inclusions (Figs. 6 and 7). It also showed that the apparently colloform growths of sphalerite and other minerals, often consist of micron- to nanometre-sized, often growth-zoned, columnar to acicular crystals, with the length of the crystals perpendicular to individual growth bands (cf. Figures 6A-C and 7B and C,

Fig. 5 Selected micrographs showing details of the Zn-Pb-rich samples presented in Fig. 3. **A)** Intergrowths of early columnar pyrite-marcasite in sample Li-HS-86. **B)** Colloform sphalerite with interlayers of calcite and associated galena overgrowing earlier pyrite and crystalline sphalerite (sample Li-HS-86). **C)** Dark-field image of the same area shown in **B)**. **D)** Early pyrite clasts with dolomite inclusions at their centre cemented by a darker sphalerite generation (Sp I), and subsequently fractured and re-cemented by a lighter colloform sphalerite (Sp II) and galena plus carbonates (sample LK1794-257.3). **E)** Dark-field image of the same area shown in **D)**. **F)** Various fossil fragments replaced by early pyrite, overgrown by later sphalerite (sample LK0179-175.6). **G)** Breccia clasts of colloform sphalerite associated with acicular galena, as well as pyrite in matrix of mostly finely fractured sulfide fragments (sample LK0644-184.4). **H)** Same as **G)** but in darkfield illumination. **I)** Pyrite clast in plastically deformed matrix of mostly sphalerite. **J)** Detail of sphalerite replacement of a clast of dolomitized limestone in sample LK1290-191.2. Sphalerite is then overgrown by pyrite. **K)** Sedimentary sphalerite infill in a cavity, covering earlier galena and pyrite grains. **L)** Darkfield image of same area as in **K)**. Mineral symbols after Warr (2021)



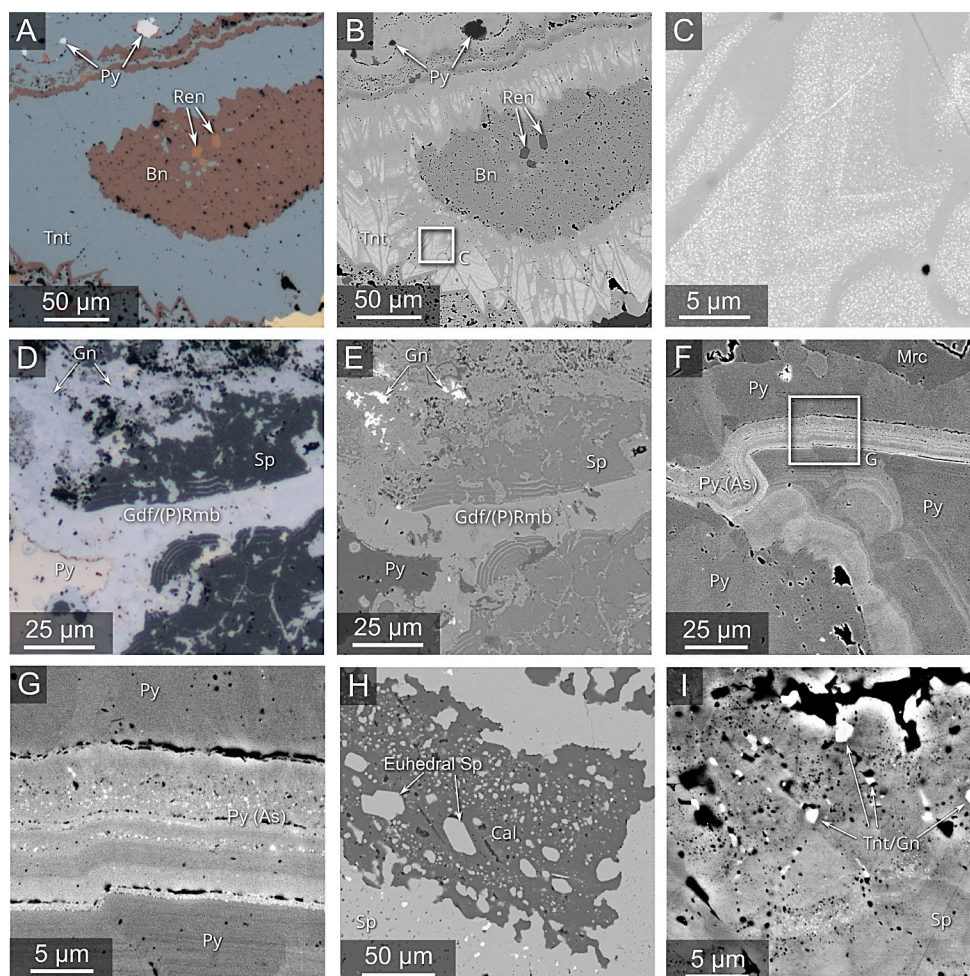
ESM Fig. A6). However, not in all cases could individual crystals be resolved (e.g., pyrite in Fig. 6G).

In sphalerite, the most abundant inclusion phases are galena and tennantite. These tend to occur along specific growth bands, giving the appearance of oscillatory Cu-Pb-zoning at larger scales (Fig. 7). In pyrite, the most abundant inclusion phases are galena, sphalerite, tennantite, and the Ni-As-minerals. The larger galena crystals themselves are relatively inclusion-free, but occasionally contain inclusions of tennantite and Pb-As sulfosalts (probably jordanite, gratonite, and/or seligmannite). Chalcopyrite and bornite often contain inclusions of tennantite, sphalerite, and galena. In

bornite, micron-sized inclusions of the Ge-mineral renierite, $(\text{Cu, Zn})_{11}(\text{Ge, As})_2\text{Fe}_4\text{S}_{16}$, are not uncommon (Fig. 6A, B). Tennantite, on the other hand, tends to contain nanoscale exsolutions of an unidentified Ag-rich phase (Fig. 6C).

The observation of abundant fine inclusions means that the measured concentrations of Cu, Pb, As, and possibly Sb in sphalerite, Zn, Pb, Cu (and sometimes Ni) in pyrite, Pb, Zn, and As in chalcopyrite, and Pb, Zn, Ge, and As in bornite, must be treated with some caution. The small sizes of some of the inclusions means that they would not have been identifiable during the examination of LA-ICP-MS ablation profiles, and will therefore not have been eliminated from

Fig. 6 Selected higher-resolution optical and back-scattered electron (BSE) images of some samples. **A)** Reflected light image of bornite with renierite inclusions in sample Li-HS-124, overgrown by tennantite crystals. **B)** Same as **A)** but BSE image. **C)** Detail of **B)** showing exsolved nano-inclusions of a Ag-rich phase in tennantite. **D)** Reflected light image of intimately associated sphalerite and gersdorffite/(para-)rammelsbergite in sample LK0985-183.4. **E)** Same as **D)** but BSE image. **F)** BSE image of an As-rich growth zone in pyrite in sample Li-HS-86. **G)** Detail of **F)** showing nano-scale inclusions of galena and other phases. **H)** BSE image of calcite containing euhedral sphalerite grains, intergrown with colloform sphalerite in sample Li-HS-86. **I)** High-resolution BSE image showing nanoscale porosity, mineral inclusions, and compositional zoning in colloform sphalerite from sample Li-HS-86



the integrated signals. Since the Ag-rich inclusions in tennantite appear to be mostly exsolutions, however, measured Ag values in tennantite are considered representative of the original tennantite composition during crystallisation.

Mineral chemistry

Overall, we conducted 1,118 LA-ICP-MS point analyses on sphalerite, pyrite, galena, chalcopyrite, bornite, and tennantite across eight samples, four from the Cu-Ni-As-rich ores, and four from the Zn-Pb-rich ores (cf. ESM Table A3). The results of these analyses are summarised in Tables 3, 4, 5, 6, and 7 in terms of the geometric means for each mineral generation in each sample. Geometric means are used, because they provide a more robust measure of central tendency for this kind of data than arithmetic means (cf. Frenzel et al. 2016b; Frenzel 2023). For the estimation of the means, values below detection limit (BDL) were replaced with the respective detection limit (cf. Frenzel 2023). Where more than 50% of all values were BDL, the means are given as <BDL. In addition to the geometric means, the results of individual point measurements including point-specific

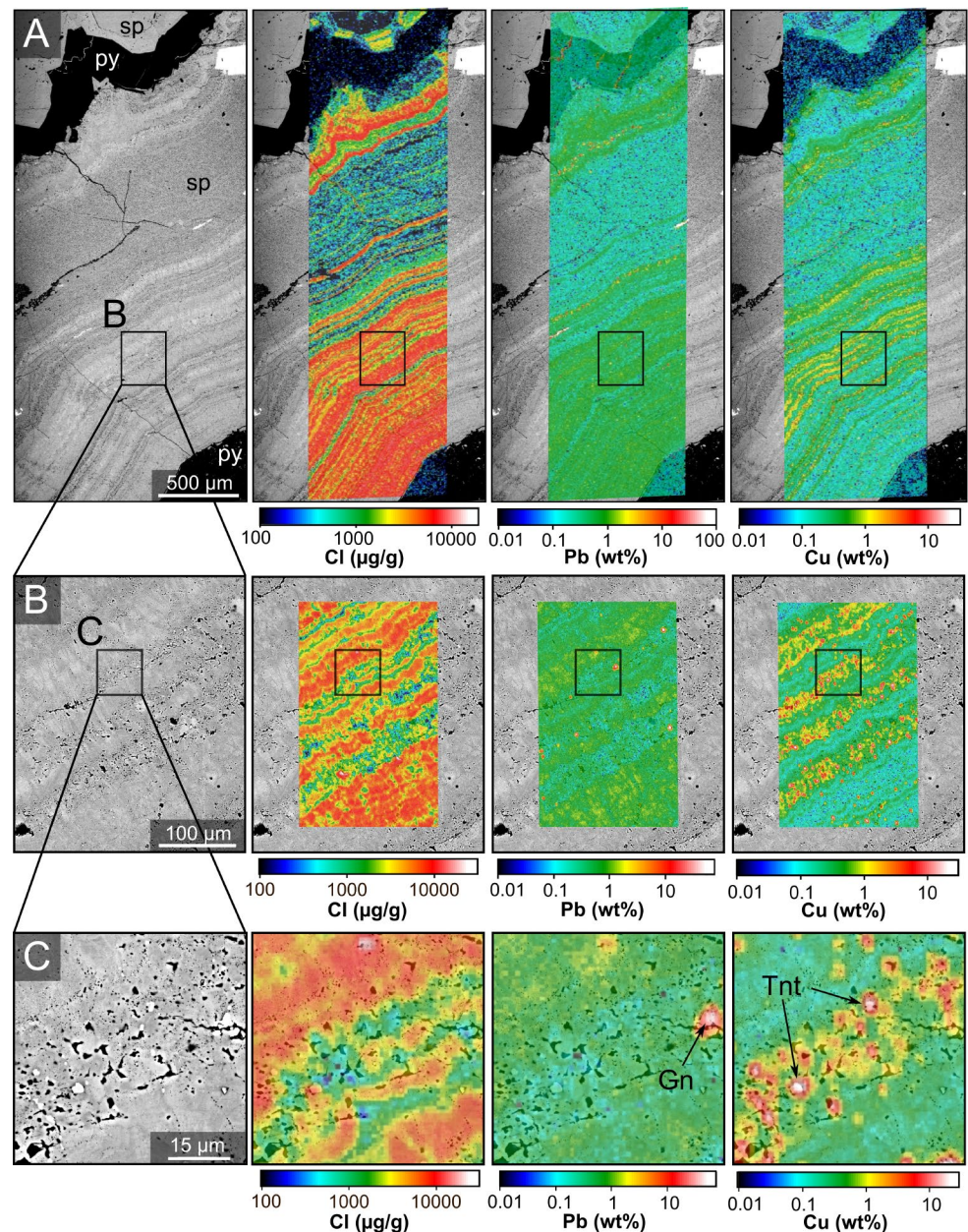
detection limits and standard errors, as well as more detailed summary statistics including 95% confidence intervals of the means, are provided in ESM Appendix B.

Note also that the mineral generation labels used in Tables 3, 4, 5, 6, and 7 are based on sample-specific sequences of mineral precipitation. As already explained above, these sequences vary considerably between samples such that correlation across samples is not possible (ESM Fig. A5). Therefore, the generation labels only indicate the *relative* order of mineral formation *within* a given sample. They do not indicate the equivalence of mineral generations *across* samples. For instance, Sp I in sample LK1114-193.1 cannot be considered equivalent to Sp I in sample Li-HS-124.

Sphalerite

The results of the sphalerite analyses are summarised in Table 3 and Fig. 8. Figure 9 and ESM Figures A7 and A8 show point transects across selected samples to illustrate the small-scale variability in sphalerite compositions within individual mineral generations. Elements not included in the

Fig. 7 Back-scattered electron (BSE) images and electron microprobe trace-element distribution maps for colloform sphalerite in sample Li-HS-86 (Sp II, cf. Figure A5 in ESM) modified from Frenzel et al. (2020) to show the distribution of Cl (as a largely lattice-hosted element), and Pb and Cu (as inclusion-hosted elements) at different scales. **A)** Coarse map recorded with a step size of 10 μm ; **B)** and **C)** high-resolution maps recorded with a step size of 1 μm . Inclusions of galena and tennantite are clearly visible in both the BSE images and elemental maps in **C)**, as are abundant nanopores and the columnar growth of individual sphalerite crystals in the BSE images in **B)** and **C)**



data displays were either below detection limit (Ni, Se, Sn, Te, Bi) in most cases, or associated with inclusions (As). As explained above, analyses for Cu, Pb, and potentially Sb may also be affected by submicron-scale inclusions but are included here for completeness and as a measure of the purity of the analysed material. Indium results are included despite being generally BDL, since they are required for the evaluation of the GGIMF is geothermometer.

Several key features are apparent. First, the ranges of trace-element concentrations in sphalerite from the Cu-rich and Zn-rich samples show considerable overlap. A two-sided t-test applied to the medians of the generation means (computed per sample) showed that the two populations are

not statistically significant for any of the investigated trace-elements at the $p=0.05$ level. The aggregation of the data to the sample level for the purpose of this test was done in accordance with the recommendations of Frenzel (2023) on dealing with hierarchical data structures.

Second, there are no consistent within-sample temporal trends¹ for most of the analysed elements at different scales, and across the dataset. For instance, Cd decreases systematically from earlier to later mineral generations in sample Li-HS-124, remains relatively constant in sample Li-HS-63,

¹ By trend, we mean a clear, systematic change in a specific direction. Random fluctuations, such as in oscillatory growth zoning, do not constitute a trend.

Table 3 Overview of sphalerite LA-ICP-MS results (geometric means)

Sample	Ore type	Generation ¹	n	Fe (wt%)	Mn (μg/g)	Co (μg/g)	Cu (μg/g)	Ga (μg/g)	Ge (μg/g)	Ag (μg/g)	Cd (μg/g)	In (μg/g)	Sb (μg/g)	Hg (μg/g) ²	Tl (μg/g)	Pb (μg/g)
LK1114-193.1	Cu	Sp I (cl)	4	0.67	43	4.5	340	3.3	24	34	7,400	<0.1	0.6	13	14	77
Li-HS-124	Cu	Sp II (cl)	5	0.57	46	<0.3	170	6.4	21	26	7,600	<0.1	<0.4	13	4.8	46
		Sp I (xl)	9	0.015	61	22	430	0.28	28	10	1,100	<0.02	<0.08	13	6.4	18
		Sp II (cl)	11	0.025	19	4.6	2,400	0.67	230	91	1,100	<0.1	0.73	30	20	150
		Sp III (cl)	10	0.009	55	22	250	<0.4	<3	4.1	930	<0.1	<0.4	26	1.5	5
Li-HS-63	Cu	Sp IV (cl)	6	0.012	46	17	170	<0.4	19	18	710	<0.1	<0.4	31	2.7	11
		Sp V (xl)	-	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
		Sp I (cl)	20	0.14	220	31	2,900	0.5	140	210	740	<0.1	12	29	590	3,800
		Sp II (cl)	29	0.23	190	27	3,200	0.4	96	160	790	<0.1	5.7	31	300	1,700
Li-HS-83	Cu	Sp III (cl)	18	0.19	320	8.6	2,800	<0.4	220	150	440	<0.1	0.6	33	530	3,000
		Sp IV (cl)	3	0.23	13	0.9	3,400	0.4	280	280	430	<0.1	1.8	21	33	540
		Sp V (cl)	13	0.15	29	2.4	1,000	<0.4	79	74	450	<0.1	0.6	46	33	220
		Sp VI (cl)	9	0.26	37	0.6	9,500	0.7	180	280	600	<0.1	0.4	39	100	1,000
Li-HS-86*	Zn	Sp VII (cl)	6	0.3	7.5	0.9	12	0.07	2.6	1.3	700	<0.02	<0.08	29	0.3	6.6
		Sp IX (cl)	2	n.d.	8.8	0.12	n.d.	1.2	210	370	770	<0.02	0.25	24	280	2,200
		Sp I (cl)	20	0.11	100	20	890	<0.4	50	190	1,500	<0.1	4.1	n.d.	64	1,600
		Sp II (cl)	10	0.14	42	18	5,900	<0.4	160	570	310	<0.1	3.8	n.d.	230	3,100
LK1794-157.3	Zn	Sp III (xl/cl)	7	0.08	62	1.7	610	<0.4	150	74	1,300	<0.1	5.1	n.d.	26	630
		Sp I (xl)	24	0.44	5.9	1.3	220	<0.4	40	16	780	<0.1	1.9	8.2	1.8	130
		Sp II (cl)	48	0.56	34	0.4	1,000	<0.4	160	130	840	<0.1	1.5	36	280	3,800
		Sp IIIa (xl)	4	0.43	250	<0.3	1,400	<0.4	180	250	910	<0.1	6	44	350	2,200
LK0644-184.4b	Zn	Sp IIIb (xl)	14	0.39	26	0.4	260	<0.06	100	7.8	560	<0.02	0.25	29	1.6	17
		Sp I (xl/cl)	19	0.09	46	7.4	3.4	2.2	54	2	270	<0.02	1	n.d.	140	840
		Sp II (cl)	15	0.04	11	3.4	1.1	0.4	28	0.91	3,400	<0.02	0.38	n.d.	140	1,400
		Sp I (cl)	39	0.25	42	0.6	640	<0.4	65	250	2,300	<0.1	0.4	6.7	1,100	2,600
LK1290-191.2	Zn	Sp I (r/xl)	7	0.08	3.6	34	18	2.1	5.8	13	630	<0.02	0.2	53	8.2	210
		Sp II (cl)	5	0.07	4.3	1.9	17	3.1	4.4	18	1,600	<0.02	0.2	45	7.3	190

n.d. – not determined

¹Specifiers in brackets indicate: cl – colloform / fine-grained sphalerite, r – replacing carbonates, xl – crystalline sphalerite (honeyblende / cleiophane)²Concentrations of Hg must be viewed as semi-quantitative due to large uncertainties regarding the Hg-content of the MASS1 standard material. However, relative differences in concentrations between samples should be reliable

*LA-ICP-MS data for Sp I and Sp II in this sample had previously been reported in Frenzel et al. (2020)

and increases in sample LK1794-157.3 (Table 3). Similarly, temporal variations of trace-element contents in the sphalerite generations within individual samples do not show any clear, systematic, and correlatable patterns across the dataset (Fig. 9; ESM Figs. A7, A8). It is noteworthy though that relatively large fluctuations may occur within generations, particularly for Ag, Co, Ga, Ge, Mn, and Tl, and also for estimated GGIMFis temperatures.

Figure 10 summarises the sphalerite formation temperatures and sulfur fugacities estimated from the LA-ICP-MS data. The numerical data, from which this figure was drafted is included in ESM Table A4. The Cu-rich and Zn-Pb-rich samples span essentially identical ranges in T and fS_2 , with average temperatures mostly between 100 and 200 °C, and average fS_2 values mostly between 10^{-12} and 10^{-20} . As also indicated in Fig. 10, however, the mineral assemblages present in some of the Cu-Ni-As-rich samples may indicate greater extremes (both high and low) in sulfidation state, than are recorded in the Zn-Pb-rich samples.

Finally, Ge contents in sphalerite range mostly from 10s to 100s of $\mu\text{g/g}$, i.e., concentrations, which could have been of commercial interest (cf. Frenzel et al. 2014). Gallium concentrations, on the other hand, are universally very low ($< 5 \mu\text{g/g}$).

Pyrite

The results of the pyrite LA-ICP-MS analyses are summarised in Table 4 and ESM Figure A9. ESM Figures A10–A12 show point transects across selected samples to illustrate the small-scale variability in pyrite compositions within individual mineral generations. Again, elements not included in the table and figures were either mostly below detection limit (Ga, Ge, In, Se, Sn, Hg, Te, and Bi), or essentially associated with inclusions (Cd). Zinc and Pb are nevertheless reported for completeness, and as a measure for the purity of the analysed material.

Several features are apparent from the pyrite data. As for sphalerite, the ranges of trace-element concentrations in pyrite from the Cu-rich and Zn-rich samples overlap greatly, and there are no statistically significant differences between the median compositions of pyrite in the two sample populations. In addition, no consistent within-sample trends are observed (Table 4, ESM Figs. A10 – A12).

The measured Co/Ni ratios fall in the range from 0.001 to 2.1, with a median value of around 0.07. Thus, most of the values are lower than the lowest values reported in the reviews by Bralía et al. (1979) and Bajwah et al. (1987) but fall well within the range of 0.01 to 2 reported for “sedimentary” pyrite in black shales (Gregory et al. 2015). While the analysed pyrite grains are virtually all hydrothermal, not sedimentary, it is worth noting that Co/Ni ratios in Lisheen

pyrite are much lower than those typically found in magmatic-related systems, such as the volcanogenic massive sulfide deposits, granite-related veins, and skarn deposits included in the datasets of Bralía et al. (1979) and Bajwah et al. (1987), broadly compatible with their formation from a non-magmatic ore fluid.

Finally, high As contents in the pyrite may be consistent with the relatively low (< 200 °C) formation temperatures indicated by sphalerite geothermometry (cf. Deditius et al. 2014). However, it is by no means clear whether the relationship reported for pyrite from precious metal deposits by Deditius et al. (2014) would hold in the present case.

Galena

The LA-ICP-MS results for galena are summarised in Table 5 and ESM Fig. A13. Note that only Ag, Cd, Sb, Tl, and Bi were above minimum detection limits in at least some of the analysed samples. However, even the mean contents of these elements are generally low, i.e., $< 100 \mu\text{g/g}$. Again, there are no statistically significant differences in galena compositions between the Cu-rich and Zn-Pb-rich samples, and the ranges in mean concentrations per generation overlap greatly. Clear within-sample temporal trends are also not apparent for any of the investigated elements. Measurable Bi only occurs in the Cu-Ni-As-rich sample LK1114-193.1, while the highest Tl contents occur in Zn-Pb-rich samples.

Overall, the trace-element contents of galena at Lisheen are low compared to those reported for other deposits by George et al. (2015). This is particularly true for the observed Ag, Sb, Bi, Se, and Te concentrations. However, the geological significance of this observation is not yet clear. We also note that George et al. (2015) did not include any samples from low-temperature carbonate-hosted Zn-Pb deposits in their study.

Chalcopyrite and bornite

Table 6 summarises the LA-ICP-MS results for chalcopyrite and bornite from the Cu-rich samples. We include these analyses here, as well as those for tennantite reported below, as a point of interest. Until now, there have been no reports of LA-ICP-MS results for Cu(Fe)-minerals from the Irish Orefield.

The main trace-elements in chalcopyrite that are probably not associated with micron- to nanometer-scale inclusions are Ag, Ge, and Tl. Bornite only hosts Ag and some Tl, but no Ge. Notably, the Ge concentrations in some of the chalcopyrite generations at Lisheen reach levels, which may potentially have been of commercial interest ($> 100 \mu\text{g/g}$), but Ga is always below detection limit. This is similar to the results seen for sphalerite above (Table 2).

Table 4 Overview of pyrite LA-ICP-MS results (geometric means)

Sample	Ore type	Generation	<i>n</i>	Mn (μg/g)	Co (μg/g)	Ni (μg/g)	Co/Ni	Cu (μg/g)	Zn (μg/g)	As (μg/g)	Ag (μg/g)	Tl (μg/g)	Pb (μg/g)
LK1114-193.1	Cu	Py I	9	2.4	450	1,800	0.25	4000	320	31,200	200	2,700	2,500
Li-HS-124	Cu	Py I	9	22	21	27	0.78	53	13	790	56	480	3.1
		Py II	8	11	300	140	2.1	240	12	1,100	42	710	86
Li-HS-63	Cu	Py III	7	12	710	1,100	0.65	1,800	52	120	210	280	6,900
		Py I	16	55	39	3,900	0.01	5,300	39	1,800	19	200	180
		Py II	15	6.3	1.5	75	0.02	100	25	11,200	1.6	2,000	690
		Py III	15	1.4	0.43	420	0.001	92	11	3,100	2.4	430	280
		Py I	19	0.6	<0.3	<2	n.d.	<2	3	4.9	<0.4	5.2	3.9
Li-HS-83	Cu	Py II	3	1.1	<0.3	12	<0.025	440	3.9	29	3.8	102	60
		Py III	4	2	0.6	25	0.024	170	6.5	5,300	10	660	110
		Py I	28	1.8	4.2	60	0.07	110	15	1,600	7.6	270	180
		Py II	7	1,200	<0.06	9.2	<0.007	150	87	980	18	160	960
		Py III	4	460	<0.3	9.5	<0.032	110	25	400	13	330	300
LK1794-157.3	Zn	Py IV	10	160	<0.3	2	<0.15	290	24	220	19	440	650
		Py V	10	38	0.25	6.1	0.041	320	12	460	11	1,200	180
		Py VI	24	62	0.22	3.4	0.065	40	11	190	2.3	590	83
		Py I	19	16	68	250	0.27	<2	15	3,000	<0.4	350	45
		Py II	7	4	4.2	66	0.064	<2	24	16,500	<0.4	3,900	55
		Py III	7	4.4	2.6	78	0.033	<2	12	12,700	<0.4	2,600	180
		Py I	27	500	7	18	0.39	82	49	1,500	18	410	400
LK-0644-184.4b	Zn	Py I	4	1.6	160	490	0.33	5.6	8.9	26,700	21	1,800	650
		Py II	11	5.7	7.4	160	0.046	4.3	12	830	12	230	1,800

Table 5 Overview of galena LA-ICP-MS results (geometric means)

Sample	Ore type	Generation	<i>n</i>	Ag (μg/g)	Cd (μg/g)	Sb (μg/g)	Tl (μg/g)	Bi (μg/g)
LK1114-193.1	Cu	Gn I	7	34	41	13	5.6	0.3
Li-HS-124	Cu	Gn I	6	20	<2.2	<0.3	1.1	<0.1
		Gn II	8	58	<2.2	<0.3	3.1	<0.1
		Gn III	5	9.5	<2.2	<0.3	1.1	<0.1
Li-HS-63	Cu	Gn I	3	51	8.1	10	23	<0.1
		Gn II	14	84	7.3	3.9	46	<0.1
		Gn III	11	34	3.6	<0.3	11	<0.1
		Gn IV	5	100	4.2	<0.3	67	<0.1
		Gn V	6	21	3	<0.3	3.1	<0.1
Li-HS-83	Cu	Gn I	10	29	7.9	9.6	4.3	<0.1
		Gn II	5	26	9.2	27	3.3	<0.1
Li-HS-86	Zn	Gn I	12	67	13	57	8.9	<0.1
		Gn II	7	110	9.2	14	15	<0.1
		Gn III	15	110	5.9	26	13	<0.1
LK1794-157.3	Zn	Gn I	5	<0.3	56	0.5	150	<0.1
		Gn II	7	<0.3	25	1.6	88	<0.1
LK-0644-184.4b	Zn	Gn I	9	77	21	14	85	<0.1
		Gn II	7	19	18	10	26	<0.1
		Gn IIIa	11	16	20	20	29	<0.1
		Gn IIIb	6	32	21	19	33	<0.1

Table 6 Overview of chalcopyrite and bornite LA-ICP-MS results (geometric means)

Sample	Ore type	Generation	<i>n</i>	Zn (μg/g)	Ga (μg/g)	Ge (μg/g)	As (μg/g)	Ag (μg/g)	Tl (μg/g)	Pb (μg/g)
LK1114-193.1	Cu	Cep I	8	91	1.2	9.6	n.d.	160	70	450
Li-HS-124	Cu	Cep II	9	100	<0.8	<6	31	21	3.9	36
		Cep I	14	<8	<0.12	37	15	23	2.6	26
		Cep II	10	<100	<0.8	43	n.d.	72	21	130
		Cep III	12	<5	<0.12	12	9.5	42	9.8	61
		Cep IV	9	110	<0.8	26	n.d.	83	69	180
		Cep V	14	<6	<0.12	14	16	67	16	85
		Cep VI	8	<7	<0.12	31	5.2	7	1.3	10
		Bn I	14	<3	<0.8	<6	<1.2	750	23	18
		Bn II	2	n.d.	<0.8	n.d.	n.d.	710	32	22
		Bn III	9	<6	1.1	<6	1.4	720	60	28
Li-HS-63	Cu	Bn IV	11	<3	<0.8	<6	4.9	810	13	16
		Bn V	5	<3	2.4	<6	<1.5	860	<0.4	6.4
		Cep I	11	<40	<0.8	570	78	51	16	100
		Cep II	12	75	<0.8	440	210	37	31	120
		Cep III	12	<50	<0.8	34	57	56	17	130
		Cep IV	4	9	<0.12	79	61	27	9.7	86
		Cep V	4	n.d.	<0.12	2.4	74	60	33	222

n.d. – not determined

Tennantite

Finally, Table 7 presents the LA-ICP-MS results for tennantite in the Cu-rich samples. These results show that tennantite is generally Zn-rich and contains only a negligible tetrahedrite (i.e., Sb) component. A variety of trace elements occurs, including Mn, Co, Ni, Ge, Ag, Cd, and Tl. The relatively high contents of Ge (up to 260 ppm),

and Tl (up to 290 ppm) in some samples are worth noting (cf. George et al. 2017), as are the uniformly high contents of Ag. Even higher Ge contents had previously been reported for tennantite-tetrahedrite mineral separates from the Tsumeb deposit, Namibia (Klunder Hansen et al. 2003). However, we are unaware of any similar reports from other deposits, or modern analyses by LA-ICP-MS showing similar contents.

Table 7 Overview of tennantite LA-ICP-MS results (geometric means)

Sample	Ore type	Generation	n	Cu (wt%)	Fe (wt%)	Zn (wt%)	Sb (wt%)	As (wt%)	S (wt%)	Mn (μg/g)	Co (μg/g)	Ni (μg/g)	Ge (μg/g)	Ag (μg/g)	Cd (μg/g)	Tl (μg/g)
LK1114-193.1	Cu	Tnt II	19	42.6	4.1	4.6	0.022	20.3	28.3	2.0	< 0.3	< 1.8	< 3.3	8300	1300	6.9
Li-HS-124	Cu	Tnt I	2	41.6	0.6	9.5	0.035	20.2	28.1	< 0.7	n.d.	n.d.	39	720	530	130
		Tnt II	12	42.2	0.2	8.9	0.029	20.2	28.2	< 2	14	< 4.4	120	610	540	210
		Tnt III	30	42.2	0.4	8.7	0.023	20.2	28.2	1.1	17	10	260	740	650	290
		Tnt IV	13	42	0.05	9.6	0.016	20.2	28.2	< 1.3	14	< 4.4	< 8.5	150	580	2.4
Li-HS-63	Cu	Tnt I	16	41.7	2.4	7.4	0.17	20	28.2	2	110	7	91	1200	350	80
		Tnt II	16	41.8	2.5	6.9	0.14	19.3	28.1	1.9	56	14	74	1400	250	90
		Tnt III	14	42.3	2.7	6.5	0.0007	20.3	28.3	2.1	< 1	< 4.9	41	830	200	6
Li-HS-83	Cu	Tnt I	48	41.8	1.3	8.5	0.023	20.2	28.2	6.7	150	53	28	2700	1100	34

Discussion

The previous section presented a range of new mineralogical and mineral-chemical data for the Cu-Ni-As- and Zn-Pb-rich ores at Lisheen. These data have several implications for the formation of Cu-rich ores in the Irish Orefield and other low-temperature carbonate-hosted Zn-Pb districts, as well as the occurrence and distribution of Ge (and Ga) concentrations that may be of economic interest. To explore these implications, the following subsections first compare our observations with previous studies of the Lisheen deposit. The genetic relationship between the Cu-Ni-As- and Zn-Pb-rich mineralisation at Lisheen is then considered in more detail, and its wider implications for the Irish Orefield are discussed. The new LA-ICP-MS data acquired in this study is used together with older sphalerite data to discuss the potential of Irish-type deposits as a potential source of the critical raw materials Ge and Ga. Finally, we briefly explore the general implications of our observations for other low-temperature carbonate-hosted Zn-Pb deposits worldwide.

Comparison to previous studies on Lisheen

The key aspects for which we must evaluate the consistency of our data against previous studies are the mineralogy and paragenesis of the ores, the chemistry of the sulfide minerals, and the inferred ore-forming conditions. The main publications, in which such data are reported and/or summarised are Hitzman et al. (2002), Fusciardi et al. (2003), Wilkinson et al. (2005a), Wilkinson (2010), Riegler and McClenaghan (2017), and Doran et al. (2022). In addition, the unpublished report of Strashimirov et al. (1999) provides mineralogical, textural, and fluid-inclusion data. Where deviations from these previous studies occur, the likely reasons are indicated.

Ore mineralogy and paragenesis

All previous studies report pyrite, marcasite, sphalerite, and galena as the major sulfide minerals at Lisheen. In addition, variable assemblages of minor minerals are reported, including the Cu minerals tennantite, chalcocopyrite, and bornite, the Ni-Co-As minerals gersdorffite, pararammelsbergite, nickeline, arsenopyrite, cobaltite, bravoite, ullmannite, and vaesite, and the Pb sulfosalts gratonite, jordanite, seligmanite, bourmonite, and boulangerite (Strashimirov et al. 1999; Hitzman et al. 2002; Fusciardi et al. 2003; Wilkinson et al. 2005a; Doran et al. 2022).

Most authors state that these minor minerals mostly occur as inclusions in galena, sphalerite, pyrite, and marcasite (cf. Strashimirov et al. 1999; Hitzman et al. 2002; Fusciardi et al. 2003; Wilkinson et al. 2005a; Doran et al. 2022). Hitzman et al. (2002) and Fusciardi et al. (2003) also report

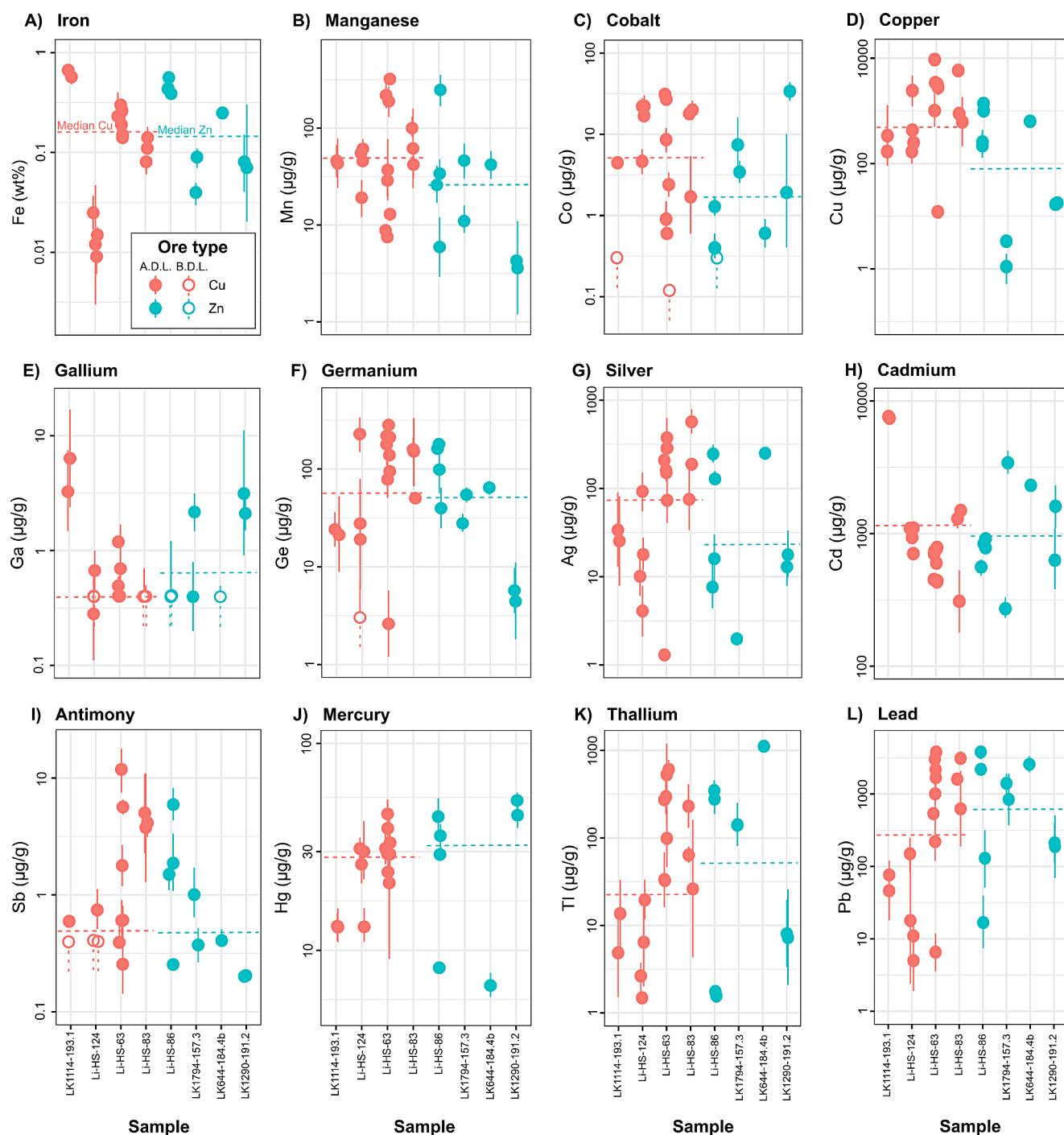


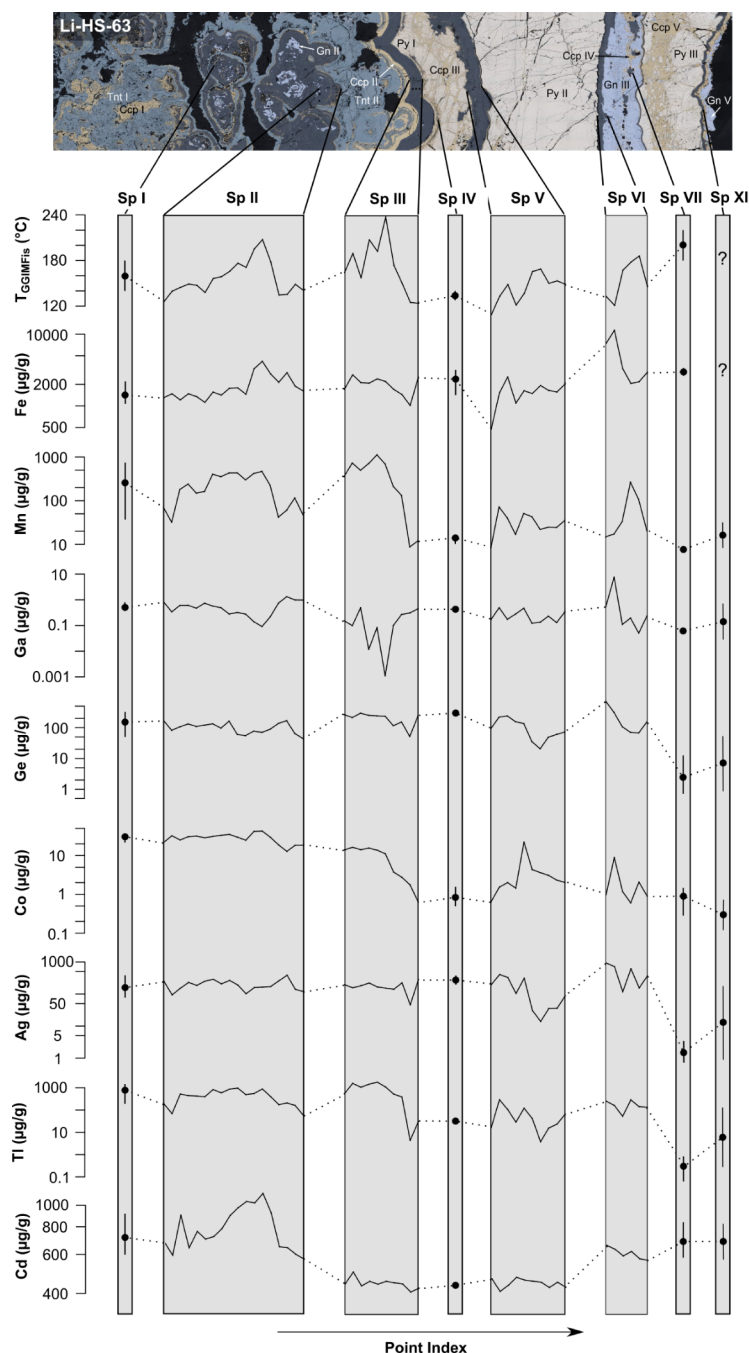
Fig. 8 Summary of sphalerite trace-element data, showing geometric means for individual mineral generations in the investigated samples, in addition to the corresponding 95% confidence intervals. The overall medians indicated by the horizontal broken lines for the sphalerites from Cu-rich and Zn-rich samples represent the medians of medi-

ans of the generation means per sample: **A)** Iron, **B)** manganese, **C)** cobalt, **D)** copper, **E)** gallium, **F)** germanium, **G)** silver, **H)** cadmium, **I)** antimony, **J)** mercury, **K)** thallium, **L)** lead. See Table 3 and ESM Appendix B for further details. A.D.L. – above detection limit; B.D.L. – below detection limit

the occurrence of coarser crystals of Cu minerals together with late honeyblende sphalerite, and Strashimirov et al. (1999) and Fuscuardi et al. (2003) note limited evidence for the occurrence of variably textured Cu-Ni-Co minerals

early during main-stage ore formation. In particular, Strashimirov et al. (1999) documents the occasional occurrence of colloform layers of Ni-As minerals intergrown with galena and pyrite-marcasite. However, all studies only describe

Fig. 9 Temporal variations of estimated GGIMFis temperatures (only best estimates) and selected trace-element concentrations in the different sphalerite generations of sample Li-HS-63. Where no clear ordering of the measurement points was possible, the median (black circle) and min-max ranges (whiskers) are shown instead of the line trace. Selected other mineral generations are labelled in accordance with ESM Fig. A5



samples in which the Cu-Ni-As minerals occur as minor or trace phases. None of the previous studies describes samples comparable to the very Cu-Ni-As-rich samples documented here (e.g., Fig. 3A-I).

The general paragenetic sequence reported for the sulfide minerals by previous authors is an early iron-sulfide rich stage, followed by the main stage of Zn-Pb ore formation, and some later post-ore carbonate veining with minor remobilisation of different sulfide minerals. The Cu-Ni-As minerals are placed either early (Fusciardi et al. 2003) or

late (Hitzman et al. 2002; Wilkinson et al. 2005) in this sequence, relative to the main ore stage.

In terms of the overall mineralogy of the ores, our observations are in good general agreement with previous studies. We also identified pyrite-marcasite, sphalerite, and galena as the major sulfides, and found that the Cu-Ni-As-rich samples generally contain abundant tennantite, chalcopyrite, bornite, gersdorffite, and (para-)rammelsbergite, and sometimes nickeline and arsenopyrite. Scarce Pb-As sulfosalt inclusions were found as inclusions in galena. In addition, vaesite occurs in one of our Zn-Pb-rich samples

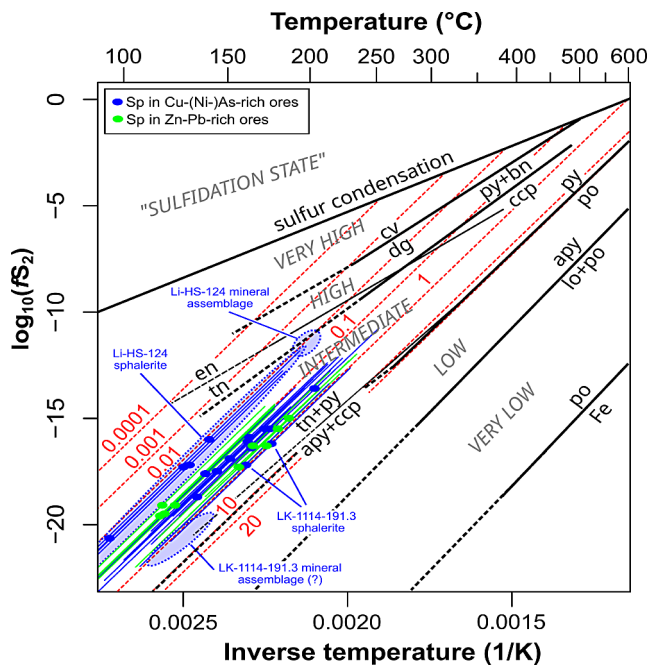


Fig. 10 Sulfur fugacity-inverse temperature plot adapted from Frenzel et al. (2022) and Einaudi et al. (2003) showing the locations of the different sphalerite generations from Lisheen relative to different mineral reactions (black lines) and isolines describing the variation of Fe contents (in mol% FeS) in sphalerite (red lines; buffered by pyrite) according to the model presented in Frenzel et al. (2022). In addition to estimates from sphalerite, tentative estimates from mineral assemblages in samples Li-HS-124 (py-bn-ccp) and LK-1114-191.3 (tn-py-apy-ccp) are included. These must however be treated with caution, since the minerals involved are not necessarily in equilibrium in these samples. Mineral abbreviations after Warr (2021)

(Li-HS-12) and we also identified minor millerite and renierite, which were not reported previously.

However, we found neither cobaltite (CoAsS) nor ullmannite (NiSbS) in our samples. We attribute this to their probable scarcity in the deposit. Gersdorffite (NiAsS) and (para)rammelsbergite (NiAs₂) are the most common Ni(-Co)-As minerals and occur in ten of our samples. Nickeline (NiAs), on the other hand, is much rarer, and was only observed in two samples, while millerite (NiS) was only seen in one. The relative rarity of cobaltite and ullmannite would be consistent with the generally low Co contents in both the ores and in pyrite (cf. Tables 1 and 4), and the low Sb contents in tennantite (cf. Table 7), which may indicate relatively low Co and Sb activities in the ore-forming system.

A key difference of our work to previous studies is that we found much more variable modes of occurrence for the Cu and Ni-As minerals, indicating their formation throughout the main stage of ore-formation. No previous study mentions the abundant occurrence of colloform tennantite, chalcopyrite, bornite, gersdorffite, (para-)rammelsbergite, and nickeline intergrown with pyrite-marcasite, sphalerite, and galena in the Cu-Ni-As-rich samples. Strashimirov et

al. (1999) and Fusciardi et al. (2003) instead mention possible evidence for the occurrence of early Cu-Ni-Co minerals in Cu-Ni-As rich samples from the southern parts of the Main Zone and Derryville Zone, which occur sometimes as thin colloform layers intergrown with pyrite or galena.

These apparent inconsistencies are easily explained by the very small number of Cu-Ni-As-rich samples apparently examined in previous studies. For instance, the descriptions of the Cu-Ni-As minerals occurring mostly as inclusions in galena, sphalerite, and pyrite-marcasite, or as late cavity and fracture fills by Hitzman et al. (2002) and Wilkinson et al. (2005a), indicate that these authors never encountered the kinds of samples described here. Strashimirov et al. (1999) also did not describe any similarly Cu-Ni-As-rich samples as those documented here.

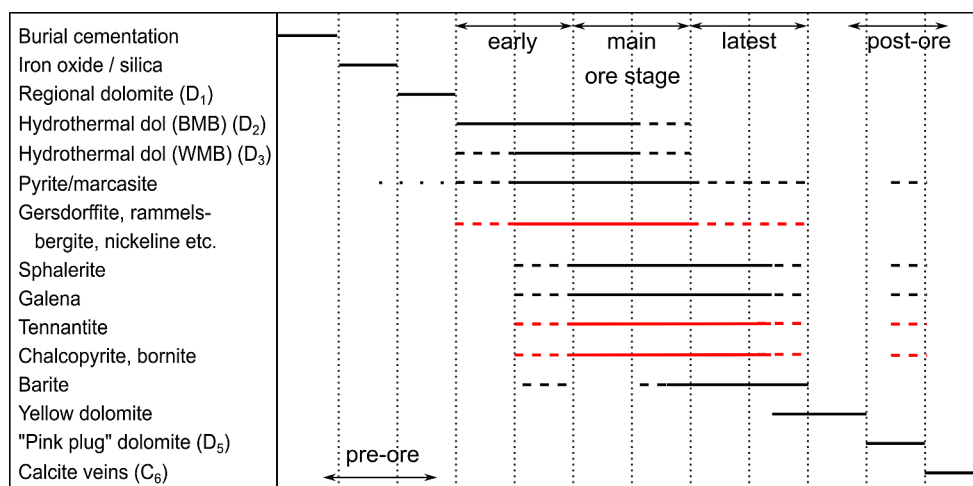
While these differences probably reflect the limited availability of sample material at the time, they mean that previous conclusions on the paragenetic positions of the Cu and Ni-As minerals at Lisheen are implicitly biased. This would explain the differences in opinion between Hitzman et al. (2002); Wilkinson et al. (2005a) on the one hand, and Fusciardi et al. (2003) and Strashimirov (1999) on the other regarding the paragenetic position of the Cu-Ni-As-rich ores.

Our observations provide clear evidence for the precipitation of Cu-Ni-As minerals throughout the main stage of ore formation. Accounting for the fact that previous studies were apparently limited by the availability of sample material, we believe this conclusion to be consistent with the overall picture presented in previous reports. A revised paragenesis diagram, updated from the most recently published one (Torremans et al. 2018), is included in Fig. 11.

Mineral chemistry

While no LA-ICP-MS results have been published for sulfide minerals from Lisheen until now, Wilkinson et al. (2005a) provided some measurement results from electron-probe micro-analysis (EPMA). These suggest appreciable mean concentrations (100s of µg/g) for a range of trace-elements in sphalerite, galena, pyrite, chalcopyrite, and tennantite. Of particular interest here are the high values of Ga and In in sphalerite, Cd, Ga, Ge, and Te in galena, Cd in pyrite, and In in tennantite reported by these authors, which are inconsistent with our results. In fact, the reported means for most of these elements appear to be calculated from relatively small numbers of observations above the detection limit, sometimes less than 1% of all measurements (e.g., 2 out of a total of 368 measurements for Mn in sphalerite; Wilkinson et al. 2005a). This means that these results are likely non-representative and biased (cf. Frenzel 2023). Furthermore, the authors do not report on the routines used to

Fig. 11 Revised paragenesis diagram for the Lisheen deposit based on Torremans et al. (2018) with modifications reflecting the observations made in this study. These modifications concern particularly the subsumption of arsenopyrite into the Gersdorffite, Rammelsbergite, Nickeline etc. category (renamed from cobaltite-nickolite to reflect the relative absence of cobaltite), the removal of tetrahedrite since none occurs in the deposit, and the introduction of bornite. Timings of the Cu-Ni-As minerals (highlighted in red) were also changed to reflect our new petrographic observations



eliminate interferences between the X-ray lines of different elements during their measurements, which may have produced additional artefacts (cf. Osbahr et al. 2015; Atanasova et al. 2015). For instance, such interferences may explain the consistently high measurements of Ge and Te in galena in Wilkinson et al. (2005a). These values cannot be correct according to our results: both elements are consistently below the detection limit ($<10 \mu\text{g/g}$) for all LA-ICP-MS measurements of galena.

Based on these considerations, we would only judge the results for Fe and Cd in sphalerite, As in pyrite, and Ag in tennantite from Wilkinson et al. (2005a) to be reliable, aside from the reported major-element concentrations. These latter results are in good general agreement with our LA-ICP-MS data, with the reported ranges showing substantial overlap.

Ore-forming conditions

Ore-forming conditions at Lisheen have previously been inferred from fluid inclusion data for hydrothermal sphalerite, calcite, dolomite, and quartz. The results are summarised in Wilkinson (2010). Measured homogenisation temperatures range from 87 to 240 °C in sphalerite, and from 87 to 277 °C in all analysed minerals, with modes occurring in the 150–210 °C range. Salinities range from 4 to 25 wt% NaCl equivalent.

The formation temperatures we inferred from sphalerite compositions using the GGIMFis geothermometer range from 94 to 203 °C for the best estimates, with most values falling in the range from 150 to 190 °C (cf. Figure 10, ESM Table A5). Including uncertainties, the overall range of possible sphalerite formation temperatures indicated by our data is ~40–250 °C (ESM Table A5). These values are in excellent agreement with the previously reported fluid inclusion analyses, particularly considering that fluid inclusion studies would only have included rare coarsely

crystalline sphalerite specimens, while we also analysed the more common colloform material. We therefore consider our sphalerite-derived temperature estimates to be reliable. No sulfur fugacities had previously been inferred for the Lisheen ores.

Genetic relationship between Cu-Ni-As-rich and Pb-Zn-rich ores

Having established the reliability of our observations, we can now proceed to consider their broader implications. As already noted above, the paragenetic relationships in our samples indicate that the Cu-Ni-As-rich mineralisation at Lisheen formed essentially at the same time as the Zn-Pb-rich mineralisation. Thus, both mineralisation types must be considered as part of the same ore-forming system, and likely formed from the same fluids, as previously suggested by Fuscuardi et al. (2003) and Torremans et al. (2018) in their explanations of the observed metal zonation patterns (Fig. 2). Textural similarities, particularly the frequent occurrence of colloform to sometimes dendritic growths for both the Zn-Pb and Cu-Ni-As minerals, further indicate that high levels of supersaturation were reached throughout the ore-forming system (cf. Barrie et al. 2009; Gagnevin et al. 2012, 2014; Luo et al. 2022) and resulted in the relatively rapid, and potentially intermittent, deposition of sulfides from the ore fluids due to variable regimes of fluid input and fluid mixing. Similar regimes have been suggested for other Irish-type deposits (e.g., Barrie et al. 2009).

Finally, the trace-element contents of sphalerite from the Cu-Ni-As- and Zn-Pb-rich ores suggest that ore formation occurred over similar, greatly overlapping ranges of temperature and sulfur fugacity for both ore types (cf. Figure 10). The only difference appears to be that more extreme values, both high and low, were reached in the Cu-Ni-As-rich ores. The similarities in the trace-element contents of pyrite and galena between the Cu-Ni-As- and Zn-Pb-rich ores further

corroborate the impression that ore-forming conditions overlapped greatly.

This overlap in ore-forming conditions may at first appear somewhat surprising. Higher mean temperatures and different mineral compositions would be expected for the Cu-rich samples from the model presented in Torremans et al. (2018) where the deep, hot, metal-bearing fluids entered the deposit from the major fault zones in the south and flowed northwards, progressively mixing with cooler sulfur-bearing fluids. However, karstic collapse events due to dissolution of the carbonate host-rocks (Wilkinson et al. 2011; Riegler and McClenaghan 2017) probably led to frequent changes in the predominant fluid pathways. Additional variability may have been introduced by variations in the overall amounts and nature of the fluid input to the system due to intermittent movements on the Lisheen Fault System (Torremans et al. 2018).

All these factors would have acted together to cause substantial local fluctuations in ore-forming conditions. There is abundant evidence for such fluctuations, the most important being considerable fluctuations of mineral compositions within individual mineral generations (e.g., Figs. 7 and 9, ESM Figs. A7, A8, A11 – A13; although we note that fine-scale growth banding in colloform minerals may also be due to boundary-layer effects and self-organisation, cf. Sun et al. 2023), frequent switches in the deposited mineral assemblages (cf. Figures 3 and 4, and ESM Fig. A2), as well as considerable fluctuations in sphalerite formation temperatures and sulfur fugacities within individual samples (up to 70 °C and 5 units of $\log_{10}(fS_2)$, cf. ESM Table A5). Any given sample thus records a relatively large range of ore-forming conditions. This noisiness in the data likely obscures the systematic mean differences expected to be present between the Cu-rich and Zn-rich ores. However, analyses of a much greater number of samples would be needed to reliably identify such differences, since they appear to be small compared to the global and local temporal fluctuations undergone by the Lisheen mineralising system.

In this context, it must also be noted that the Cu-Ni-As minerals within any given sample generally formed separately from (but often interspersed with) the sphalerite. Thus, somewhat different conditions may have prevailed during their precipitation. However, these differences are not expected to be substantial, as indicated by samples with co-deposited sphalerite and Cu-Ni-As-minerals. For instance, Sp I in sample Li-HS-63 was deposited together with chalcopyrite and tennantite and records an average formation temperature of 161 ± 50 °C (ESM Table A5), while Sp II in sample Li-HS-124 appears to have formed together with bornite and indicates an average formation temperature of 140 ± 53 °C.

To summarise, different lines of evidence suggest that the Cu-Ni-As-rich ores at Lisheen formed from the same fluids as the Zn-Pb-rich ores, as had previously been suggested (Fusciardi et al. 2003; Torremans et al. 2018). However, while some previous authors had suggested that the occurrence of the Cu-Ni-As minerals may consistently indicate relatively high formation temperatures for the Cu-rich ores, i.e., above 250 °C (Fusciardi et al. 2003; Wilkinson et al. 2005a), this is not corroborated by our data, which indicates similar T and fS_2 conditions for the Cu-rich and Zn-rich ores, with temperatures < 250 °C.

External data from stratiform sediment-hosted Cu deposits (Hitzman et al. 2005; Poetz et al. 2022), as well as five-element veins (Guilcher et al. 2021), does in fact show that low- to moderate-temperature fluids (< 100–250 °C) can transport and deposit significant quantities of Cu, Ni, Co, and As. While the ore fluids in these deposit types may have been more oxidised than Irish-type fluids (cf. Wilkinson 2023), enhancing their ability to transport Cu, Ni, Co, and As in solution, hydrothermal leaching experiments on potential source rocks for Irish-type deposits (Bischoff et al. 1981) do in fact indicate the ability of the principal ore fluids to mobilise significant Cu, Ni, and As in addition to Zn and Pb at temperatures around 200 °C (cf. discussion on metal contents in ores below; Table 8). Therefore, the high fluid temperatures suggested by some authors to explain the Cu-Ni-As-rich mineralisation do not appear to be strictly necessary.

Cu-rich ores in the Irish Orefield

The contemporaneous formation of Cu-Ni-As-rich and Zn-Pb-rich mineralisation at Lisheen has important implications for the interpretation and exploration of Cu-rich deposits and prospects elsewhere in the Irish Orefield. Specifically, it strongly supports the suggestion that deposits like Gortdrum or Tullacondra form the Cu-rich root zones of larger ore-forming systems for which the Zn-Pb-rich parts have been eroded away (Steed 1986; Dunlevy et al. 2023). Alternatively, our observations could also support the hypothesis of the contemporaneous formation of Irish-type deposits with different metal contents due to large-scale regional fluid flow and metal fractionation from SW to NE, as proposed by Johnston et al. (1996) and Johnston (1999). However, this latter hypothesis is at odds with the local sourcing of Pb from the basement rocks underlying individual deposits, as demonstrated by isotopic studies (LeHuray et al. 1987; Everett et al. 2003).

In addition, our observations render several other hypotheses recently put forward by Cordeiro et al. (2023) and Silva et al. (2022) less likely. These authors had assumed that Cu-rich mineralisation in the Irish Orefield generally occurs as

Table 8 Comparison between Lisheen deposit and suggested ore fluid compositions

Metal ratio	F.I. data from basement veins ¹	Brine reacted with greywacke ²	F.I. data from Silvermines ³	Lisheen deposit ⁴	Silvermines deposit ⁶	Kilbricken prospect ⁵	Tynagh deposit ⁶	Tullacondra prospect ⁷	Gortdrum deposit ⁶
Ag/Zn (10^{-4} g/g)	n.d.	n.d.	n.d.	2.3	3.6	10	12	2,700	> 250
Cu/Zn (10^{-4} g/g)	1,200–5,000	1,000–1,200	n.d.	55	n.d.	650	890	670,000	> 120,000
Ni/Zn (10^{-4} g/g)	n.d.	1,800–2,700	n.d.	46	n.d.	n.d.	n.d.	1,800	n.d.
As/Zn (10^{-4} g/g)	n.d.	800–830	n.d.	220	n.d.	2,800	n.d.	7,100	> 1,000
Fe/Zn (g/g)	0.8–3.1	23–100	n.d.	1.7	n.d.	2.1	n.d.	150	n.d.
Pb/Zn (g/g)	1.9–5.8	0.02–0.09	0.1–1.0	0.17	0.39	0.6	1.1	0.06	n.d.

n.d. – not determined

¹Wilkinson et al. (2005b), reported range represents the range of the values obtained on individual samples

²Bischoff et al. (1981), reported range represents the results of experiments at 200 and 350 °C, respectively

³Wilkinson et al. (2009), reported range represents that inferred from thermodynamic calculations, based on measured Pb concentrations in inclusions

⁴Calculated from mine assay database (bdl values replaced by detection limits)

⁵Colthurst and Reed (2019), values for Zn, Pb, and Cu and corresponding ratios are taken from the resource estimate while As/Zn and Fe/Zn were estimated from assay data included in the report

⁶Ashton et al. (2023), for Gortdrum, a maximum Zn content of 0.1 wt%, and minimum As content of 0.01 wt% was assumed (cf. Steed 1986; Cordeiro et al. 2023)

⁷Silva et al. (2022), values estimated from analyses for samples with > 0.1 wt% Cu

a later, separate event that overprints the Zn-Pb-rich ores, citing Lisheen, in particular, as a prominent example of such a supposed relationship.

Another important implication of the suggested contemporaneous nature of Cu-Ni-As- and Zn-Pb-rich mineralisation in Ireland concerns the expected abundance of Cu-rich deposits, prospects, and orebodies relative to classic Zn-Pb-rich Irish-type deposits. To evaluate this, it is instructive to examine the probable compositions of the principal ore fluids and compare them to the compositions of individual deposits. This is done in Table 8. Fluid data are derived from fluid inclusions in basement-hosted quartz-carbonate-sulfide veins, inferred to be potential feeder structures for the stratigraphically higher mineralisation (Wilkinson et al. 2005b), from fluid inclusions in sphalerite from the Silvermines deposit (Wilkinson et al. 2009), and from hydrothermal leaching experiments conducted on Ordovician greywacke from the southwestern part of the Irish midlands (Bischoff et al. 1981). Of the deposits included in Table 8, Lisheen and Silvermines are representative of the classic Zn-Pb-rich Irish type deposits, while Tynagh and the Kilbricken prospect contain mixed Cu-Zn-Pb ores, and the Tullacondra prospect and Gortdrum deposit are virtually Zn-Pb-free. Data are presented in terms of element-to-zinc ratios so that comparisons can easily be made between the fluids and ores.

It is apparent from the comparison between deposits and fluids that much less Cu, Ni, and As is present (relative to Zn) in the classic Zn-Pb-rich Lisheen deposit than would be expected if the entire metal load of the pristine ore

fluids had been precipitated within the deposit. On the other hand, Kilbricken and Tynagh have Cu/Zn and As/Zn ratios approaching or exceeding the expected ones, and Tullacondra and Gortdrum far exceed expected ratios. Tullacondra also shows a similar Ni/Zn ratio to the expected ore fluids. Unfortunately, no further information is available for Ni in other deposits.

The fact that Cu/Zn and As/Zn ratios at Tullacondra and Gortdrum exceed expected ratios provides further support for the interpretation that these deposits represent Cu-rich core zones remaining after erosion of the stratigraphically higher, more Zn-Pb-rich parts of the orebodies. Kilbricken and Tynagh, in contrast, may represent more or less completely preserved deposits, while the absence of Cu, Ni, and As from Lisheen and other deposits must have other reasons.

Given that Cu and Ni would be expected to be much less mobile than Zn and Pb during the fluid-mixing process inferred as the main ore-forming mechanism in most Irish-type deposits (Frenzel et al. 2019), their relative absence from classic Irish-type deposits such as Lisheen can only be explained by their absence from the metal-bearing fluids that formed the *known* orebodies. The easiest explanation for this absence, assuming that pristine ore fluids in all systems were similar to the compositions reported in Wilkinson et al. (2005b) and Bischoff et al. (1981), would be the precipitation of considerable amounts of Cu-Ni-As-rich mineralisation at depth before the fluids reached the main site of Zn-Pb ore formation. In the case of Lisheen, such mineralisation would need to contain about ~300 kt of each

Cu and Ni metal, and ~200 kt of As to make up the balance “missing” from the deposit.

Two modes of occurrence can be envisaged for such mineralisation: (1) it may occur in discrete orebodies reaching economic grades at stratigraphically lower levels than the major Zn-Pb mineralisation, even including the Old Red Sandstone, such as at Kilbricken, Tynagh, Gortdrum, Tullacondra, and Aherlow, or (2) it may be disseminated at subeconomic grades over relatively large rock volumes at depth. Unfortunately, only very little deep drilling has been done at Lisheen and other Irish-type deposits to explore for either of these two mineralisation types. We note, however, that both disseminated and vein-hosted Cu-mineralisation occur in the Old Red Sandstone in the Munster Basin in SW Ireland. Furthermore, these are interpreted to have formed in similar settings, under similar conditions, and around the same time, as the carbonate-hosted deposits of the Irish Midlands (Kinnaird et al. 2002; Lang et al. 2020).

Experiments, in which metal-bearing fluids were reacted with samples of Old Red Sandstone do in fact indicate the quantitative precipitation of Cu and Co during these reactions (Wilkinson 2023). However, the same is not true for Ni, and no results are reported for As. While bleached zones within the Old Red Sandstone are not common underneath the Lisheen deposit (cf. Wilkinson 2023), it cannot be excluded that such a mechanism contributed to the apparent lack of Cu, Ni, and As in many deposits. In addition, quartz-carbonate-sulfide veins, which often contain some Cu minerals, occur throughout the Paleozoic basement rocks in the Irish Midlands (Wilkinson et al. 2005b), and may have provided another sink for these elements.

An alternative explanation for the occurrence of ore fluids with less Cu, Ni, and As could be regional variations in the average composition and/or leachability of the main metal source rocks. Some data on such variability is presented for hand-specimen-sized samples in Wilkinson (2023). These data do indeed show some variability in the relative leachability of the different metals (Zn, Pb, Cu, Ni, As etc.). However, the observed differences do not reach several orders of magnitude, as would be required to explain the observed differences in metal contents within the deposits (Table 8). Furthermore, the leached rock volumes feeding individual ore deposits are expected to be at least six orders of magnitude larger than the investigated samples, reducing the expected extent of variability.

Whatever the true reason for the relative absence of Cu-Ni-As-rich ores from the Irish Orefield, it is certainly true that past exploration activities in the Irish Midlands mostly focused on the classic Waulsortian-hosted Zn-Pb deposits (Ashton et al. 2023). Thus, potential Cu-Ni-As-rich mineralisation existing at lower stratigraphic levels may be underexplored. The comparison between currently inferred ore

fluid compositions (mean Cu/Zn ratios of ~0.1) with identified resources, which are mostly Zn-Pb-rich, suggests that there may be significant potential for additional discoveries.

If the simple assumption is made that potentially economic Cu mineralisation with grades of ~1% Cu formed just as frequently as economic Zn-Pb mineralisation, then the expected total tonnage of Cu ores in the Irish Orefield should be similar to that of the Zn-Pb ores, i.e., greater than 200 Mt (cf. Ashton et al. 2023). At present, <30 Mt of such resources are known (Ashton et al. 2023). The most likely locations for Cu-rich ores to occur would be near major Zn-Pb deposits, but generally in lower stratigraphic positions, reaching as deep as the Old Red Sandstone.

Ge (and Ga) potential of Irish base-metal ores

The LA-ICP-MS analyses presented in this article show that sphalerite at Lisheen contains Ge concentrations in the range from 10s to 100s of µg/g, while Ga concentrations are generally below 5 µg/g. Concentrations above 100 µg/g of Ga or Ge in sphalerite (=zinc concentrate) are generally sufficient for these elements to be of commercial interest for by-product recovery during zinc refining (cf. Frenzel et al. 2014, 2016a). Our analyses suggest that the Zn-Pb ores at Lisheen would probably have reached such values for Ge, but certainly not Ga. It is currently unclear whether Ge extraction actually happened at the refineries the Zn concentrates were shipped to.

Since the Lisheen deposit is already mined out, the observations from this study can only serve as a rough indicator for the potential of other deposits in the Irish Orefield. Currently, additional data on Ga and Ge in sphalerite for Irish-type deposits are only available for the Keel Prospect (Watling 1976), where average concentrations are 59 µg/g Ga and 14 µg/g Ge. One sample from Ballyhickey included in the study of Shazly et al. (1957) yielded no measurable Ga or Ge (<3 µg/g each), and one sample from County Wicklow yielded 70 µg/g Ga and 400 µg/g Ge (Shazly et al. 1957). While these results are quite promising for the Irish Orefield in general, the relatively large variability between samples and deposits means that new analyses will generally be needed to determine the suitability of any given deposit or prospect as a source of Ge and/or Ga-rich Zn concentrates.

Perhaps of greater interest than the analytical results for sphalerite are our measured Ge concentrations in chalcopyrite and tennantite, as well as the occurrence of the Ge-mineral renierite, documented in this study. These observations indicate that Cu-rich Irish-type ores could also be potential sources of Ge, similar to the Cu-rich orebodies at Kipushi and Tsumeb (Höll et al. 2007; Belissont et al. 2019). This is particularly interesting in combination with the possible

exploration potential of the Cu-rich ores noted above. However, substantial variability in the Ge contents between localities is expected, and individual mineralisations will therefore still need to be evaluated on a case-by-case basis.

Implications for other MVT districts

The implications of our observations extend beyond the Irish Orefield. If significant exploration potential exists for Cu(-Ni)-rich ores in Ireland, it may also be present in other low-temperature carbonate-hosted Zn-Pb districts around the world. But what determines this potential?

In general, Cu-rich ores in MVT-type deposits are rare (cf. Leach et al. 2005). However, Zn-Pb mineralisation commonly occurs in association with sediment-hosted stratiform Cu (SSC) deposits, even though it is usually subeconomic in this setting (Oszczepalski 1999; Hitzman et al. 2005). Indeed, the metal-bearing fluids in SSC and MVT deposits are thought to be quite similar: high-salinity (> 10 wt% NaCl-eqv), low-temperature (< 200 °C) brines. The major difference is that these brines are thought to be oxidised in SSC deposits, while they can be either oxidised (hematite-buffered) or reduced in MVT deposits. A further difference is the source-rock for the metals: in SSC deposits, this is generally thought to be a red-bed sequence, potentially intercalated with mafic to intermediate volcanics or magmatic rocks (cf. Hitzman et al. 2005), while MVT-type deposits can have more variable metal sources, often including older basement rocks. Trapping mechanisms in both deposit types are similar, involving either the reduction of oxidised sulfur present in the fluids and/or the addition of reduced sulfur.

Based on this comparison, we suggest that the absence of Cu in many MVT districts may be due to both the absence of suitable source rocks in the deeper parts of the hydrothermal system, as well as the oxidation state of the metal-bearing fluids. It is probably more difficult to mobilise significant amounts of Cu in a reducing than an oxidising fluid at the relatively low temperatures prevalent in these mineral systems (cf. Haynes and Bloom 1987). MVT districts where the metal-bearing fluids were more reducing and had lower temperatures, and where suitable source rocks for Cu are absent, may therefore be relatively Cu-poor. This would imply the following criteria for the general determination of Cu exploration potential in MVT-type Zn-Pb districts:

- 1) Are the metal-bearing fluids expected to be mostly oxidised or reduced?
- 2) Did the metal-bearing fluids reach temperatures in excess of 200 °C?

- 3) Are suitable source-rocks (e.g., continental red beds or mafic to intermediate magmatic rocks) present, from which Cu could have been mobilised?

In addition, it is a positive sign if some Cu-rich mineralisation has already been recognised in the district, such as in the Irish Midlands. If doubtful, the ability of potential metal source rocks to contribute sufficient Cu to the mineralising brines under relevant leaching conditions can be tested by targeted experiments (cf. Bischoff et al. 1981).

If indications for Cu potential exist in the district, the general expectation would be for the Cu-rich ore bodies to be located stratigraphically lower / deeper than, but in close spatial association with, the Zn-Pb mineralisation, potentially controlled by the same structures. Alternatively, the Cu-rich mineralisation may occur as disseminations.

Conclusions

In this contribution, we provided detailed evidence for the contemporaneous formation of Cu-Ni-As- and Zn-Pb-rich ores at the classic Lisheen Zn-Pb deposit, Ireland. We also showed that both ore types may have had the potential for commercial Ge extraction.

These observations have important implications for the Irish Orefield and other low-temperature carbonate-hosted Zn-Pb districts. First, the cogenetic nature of the Cu-Ni-As-rich and Zn-Pb-rich ores, the relative absence of Cu-Ni-As-rich ores from most of the larger Zn-Pb deposits compared with expectations derived from probable ore fluid compositions, and finally, the known characteristics of isolated Cu-Ni-As-rich ore bodies like Gortdrum and Tullacondra, indicate that significant Cu-Ni-As-rich mineralisation may be present at lower stratigraphic levels across the Irish Orefield. Areas with known Zn-Pb mineralisation may be particularly prospective for such ores, which may occur as deep as the Old Red Sandstone.

Second, there may be considerable exploration potential for Cu-rich mineralisation in other MVT-type Zn-Pb districts. Consideration of major controls on the metal budgets in such deposits suggests that districts, in which the metal bearing fluids were oxidised, reached higher temperatures (> 200 °C), and source rocks contained leachable Cu, may be particularly prospective. The Cu-rich mineralisation would generally be expected to occur at deeper levels than the Zn-Pb mineralisation.

Finally, sparse evidence on mineral compositions from Lisheen, the Keel prospect, and isolated analyses reported in Shazly et al. (1957) indicate that Irish-type ores may generally be prospective as potential sources of mineral concentrates for by-product Ge and Ga production. However,

the large variability between samples and deposits means that this potential must be assessed on a case-by-case basis, and cannot easily be guaranteed or generalised.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00126-024-01299-8>.

Acknowledgements This research was supported by a P.R.I.M.E. Fellowship awarded to MF by the German Academic Exchange Service (DAAD) and completed in part at the University of Adelaide, South Australia, and the TU Bergakademie Freiberg, Germany. Funding for the initial field work in 2015, the preparation of the samples, and the completion of two M.Sc. theses (MR and Dominik Schlunkert) was generously provided by the Helmholtz Institute Freiberg for Resource Technology on the initiative of Jens Gutzmer, who suggested this research topic to the first author. The authors are also greatly indebted to Vedanta Zinc International for granting us access to their drill-core database, and supporting our sampling from the underground workings, core store, and ore stockpiles. The preparation team at the Helmholtz Institute Freiberg for Resource Technology is thanked for the beautiful sections they prepared for this study. Dominik Schlunkert assisted with sampling and provided some of the initial sample descriptions as part of his M.Sc. thesis.

Funding Open Access funding enabled and organized by Projekt DEAL.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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References

- Andrew CJ (1986) The tectono-stratigraphic controls to mineralization in the Silvermines area, County Tipperary, Ireland. In: Andrew CJ et al (eds) *Geology and genesis of mineral deposits in Ireland*. Irish Association for Economic Geology, Dublin, p 377–418
- Ashton JH, Andrew CJ, Hitzman (2023) Irish-type Zn-Pb deposits – what are they and can we find more? In: Andrew CJ et al (eds) *Irish-type deposits around the world*. Irish Association for Economic Geology, Dublin, pp 95–146
- Atanasova P, Krause J, Möckel R, Osbahr I, Gutzmer J (2015) Electron probe microanalysis of REE in eudialyte group minerals: challenges and solutions. *Microsc Microanal* 21:1096–1113
- Bajwah ZU, Seccombe PK, Offler R (1987) Trace element distribution, Co:Ni ratios and genesis of the big Cadia iron-copper deposit, New South Wales, Australia. *Min Deposita* 22:292–300
- Balabin AI, Urusov VS (1995) Recalibration of the sphalerite cosmo-barometer: experimental and theoretical treatment. *Geochim Cosmochim Acta* 59:1401–1410
- Barrie CD, Boyce AJ, Boyle AP, Williams PJ, Blake K et al (2009) On the growth of colloform textures: a case study of sphalerite from the Galmoy ore body, Ireland. *J Geol Soc Lond* 166:563–582
- Barton PB, Toulmin P (1966) Phase relations involving sphalerite in the Fe-Zn-S system. *Econ Geol* 61:815–849
- Bauer ME, Burisch M, Ostendorf J, Krause J, Frenzel M, Seifert T, Gutzmer J (2019) Trace element geochemistry of sphalerite in contrasting hydrothermal systems of the Freiberg District, Germany: insights from LA-ICP-MS analysis, near-infrared light microthermometry of sphalerite-hosted fluid inclusions, and sulfur isotope geochemistry. *Min Deposita* 54:237–262
- Belissont R, Munoz M, Boiron M-C, Luais B, Mathon O (2019) Germanium crystal chemistry in Cu-bearing sulfides from micro-XRF mapping and micro-XANES spectroscopy. *Minerals* 9:227
- Bischoff JL, Radtke AS, Rosenbauer RJ (1981) Hydrothermal alteration of graywacke by brine and seawater: roles of alteration and chloride complexing on metal solubilization at 200° and 350°C. *Econ Geol* 76:659–676
- Bralia A, Sabatini G, Troja F (1979) A revaluation of the Co/Ni ratio in pyrite as geochemical tool in ore genesis problems – evidences from Southern Tuscany pyritic deposits. *Min Deposita* 14:353–374
- Cavender BD, Shelton KL, Schiffbauer JD (2016) An atypical orebody in the Brushy Creek Mine, Viburnum Trend, MVT District, Missouri: early Cu-(Ni-Co)-Zn-rich ores at the Lamotte Sandstone/Bonnetterre Dolomite contact. *Econ Geol* 111:259–269
- Colthurst J, Reed G (2019) National Instrument 43–101 technical report (amended and restated) on the mineral resource estimate for the Kilbricken zinc-silver-lead-copper project Co. Clare, Ireland for Hannan Metals Ltd., amended and restated 28th January 2019. 143 p
- Cordeiro P, Matias dos Santos A, Steed G, de Araújo Silva A, Meere P et al (2023) The carbonate-hosted Gortdrum Cu-Ag-(Sb-Hg) deposit, SW Ireland: C-O-Sr-Nd isotopes and whole-rock geochemical signatures. *J Geochem Explor* 248:107196
- Deditius AP, Reich M, Kesler SE, Utsunomiya S, Chrysosoulis SL et al (2014) The coupled geochemistry of Au and As in pyrite from hydrothermal ore deposits. *Geochim Cosmochim Acta* 140:644–670
- Derry DR, Clark GR, Gillatt N (1965) The Northgate base-metal deposit at Tynagh, County Galway, Ireland. *Econ Geol* 60:1218–1237
- Dmitrijeva M, Metcalfe AV, Ciobanu CL, Cook NJ, Frenzel M et al (2018) Discrimination and variance structure of trace element signatures in Fe-oxides: a case study of BIF-mineralisation from the Middleback Ranges, South Australia. *Math Geosci* 50:381–415
- Doran AL, Hollis SP, Menuge JF, Piercey SJ, Boyce AJ et al (2022) A distal, high-grade Irish-type orebody: Petrographic, sulfur isotope, and sulfide chemistry of the Island Pod Zn-Pb orebody, Lisheen, Ireland. *Econ Geol* 117:305–326
- Dunlevy E, Torremans K, Holdstock M, Hitzman M (2023) A new look at the Gortdrum copper deposit, Co. Tipperary, Ireland. In: Andrew CJ et al (eds) *Irish-type deposits around the world*. Irish Association for Economic Geology, Dublin, pp 363–370
- Einaudi MT, Hedenquist JW, Inan EE (2003) Sulfidation state of fluids in active and extinct hydrothermal systems: transition from porphyry to epithermal environments. In: Simmons SF, Graham I (eds) *Volcanic, geothermal, and ore-forming fluids: rulers and witnesses of processes within the Earth*, SEG Special Publication, Society of Economic Geologists, Littleton, Co., vol. 10, p 285–313

- El Shazly EM, Webb JS, Williams D (1957) Trace elements in sphalerite, galena and associated minerals from the British Isles. *Trans Inst Min Met* 66:241–271
- Everett CE, Rye DM, Ellam RM (2003) Source or sink? An assessment of the role of the Old Red Sandstone in the genesis of the Irish Zn-Pb deposits. *Econ Geol* 98:31–50
- Frenzel M (2023) Making sense of mineral trace-element data – how to avoid common pitfalls in statistical analysis and interpretation. *Ore Geol Rev* 159:105566
- Frenzel M, Ketris MP, Gutzmer J (2014) On the geological availability of germanium. *Min Deposita* 49:471–486
- Frenzel M, Ketris MP, Seifert T, Gutzmer J (2016a) On the current and future availability of gallium. *Resour Policy* 47:38–50
- Frenzel M, Hirsch T, Gutzmer J (2016b) Gallium, germanium, indium, and other minor and trace elements in sphalerite as a function of deposit type – A meta-analysis. *Ore Geol Rev* 76:52–78
- Frenzel M, Mikolajczak C, Gutzmer J, Reuter MA (2017) Quantifying the relative availability of high-tech by-product metals – the cases of gallium, germanium and indium. *Resour Policy* 52:327–335
- Frenzel M, Röhner M, Gutzmer J, Burisch M, Cook NJ et al (2019) Explaining metal zonation at the Lisheen Zn-Pb deposit. In: *Proceedings, 15th SGA Biennial Meeting on Life with Ore Deposits on Earth*, Glasgow, pp 214–217
- Frenzel M, Cook NJ, Ciobanu CL, Slattery AD, Wade BP et al (2020) Halogens in hydrothermal sphalerite record origin of ore-forming fluids. *Geology* 48:766–770
- Frenzel M, Voudouris P, Cook NJ, Ciobanu CL, Gilbert S, Wade BP (2022) Evolution of a hydrothermal ore-forming system recorded by sulfide mineral chemistry: a case study from the Plaka Pb-Zn-Ag deposit, Lavrion, Greece. *Min Deposita* 57:417–438
- Fusciardi LP, Güven JF, Stewart DRA, Carboni V, Walsh JJ (2003) The geology and genesis of the Lisheen Zn-Pb deposit, Co. Tipperary, Ireland. In: Kelly JG et al (eds) *Europe's Major Base Metal deposits*. Irish Association for Economic Geology, Dublin, pp 455–482
- Gagnevin D, Boyce AJ, Barrie CD, Menuge JF, Blakeman RJ (2012) Zn, Fe and S isotope fractionation in a large hydrothermal system. *Geochim Cosmochim Acta* 88:183–198
- Gagnevin D, Menuge JF, Kronz A, Barrie C, Boyce AJ (2014) Minor elements in layered sphalerite as a record of fluid origin, mixing, and crystallization in the Navan Zn-Pb ore deposit, Ireland. *Econ Geol* 109:1513–1528
- George L, Cook NJ, Ciobanu CL, Wade BP (2015) Trace and minor elements in galena: a reconnaissance LA-ICP-MS study. *Am Mineral* 100:548–569
- George LL, Cook NJ, Ciobanu CL (2017) Minor and trace elements in natural tetrahedrite-tennantite: effects on element partitioning among base metal sulphides. *Minerals* 7:17
- Godefroy-Rodriguez M, Hagemann S, Frenzel M, Evans NJ (2020) Laser ablation ICP-MS trace element systematics of hydrothermal pyrite in gold deposits of the Kalgoorlie district, Western Australia. *Min Deposita* 55:823–844
- Gregory DD, Large RR, Halpin JA, Baturina EL, Lyons TW, Wu S et al (2015) Trace element content of sedimentary pyrite in black shales. *Econ Geol* 110:1389–1410
- Guilcher M, Schmaucks A, Krause J, Markl G, Gutzmer J, Burisch M (2021) Vertical zoning in hydrothermal U-Ag-Bi-Co-Ni-As systems: a case study from the Annaberg-Buchholz District, Erzgebirge (Germany). *Econ Geol* 116:1893–1915
- Güven J, Torremans K, Johnson S, Hitzman M (2023) The Rathdowney Trend: geological evolution and controls on Zn-Pb mineralisation. In: Andrew CJ et al (eds) *Irish-type deposits around the world*. Irish Association for Economic Geology, Dublin, pp 329–362
- Haynes DW, Bloom MS (1987) Stratiform copper deposits hosted by low-energy sediments: III. Aspects of metal transport. *Econ Geol* 82:635–648
- Hitzman MW, Redmond PB, Beaty DW (2002) The carbonate-hosted Lisheen Zn-Pb-Ag deposit, County Tipperary, Ireland. *Econ Geol* 97:1627–1655
- Hitzman MW, Beaty DW (2003) The Irish Zn-Pb-(Ba-Ag) Orefield. In: Kelly JG et al (eds) *Europe's Major Base Metal deposits*. Irish Association for Economic Geology, Dublin, pp 499–531
- Hitzman MW, Kirkham R, Broughton D, Thorson J, Selley D (2005) The sediment-hosted stratiform copper ore system. In: Hedenquist JW (eds) *One Hundredth Anniversary Volume*, Society of Economic Geologists, Golden, CO, pp 609–942
- Hnatyshin D, Creaser RA, Meffre S, Stern RA, Wilkinson JJ, Turner EC (2020) Understanding the microscale spatial distribution and mineralogical residency of Re in pyrite: examples from carbonate-hosted Zn-Pb ores and implications for Re-Os geochronology. *Chem Geol* 533:119427
- Hnatyshin D, Creaser RA, Wilkinson JJ, Gleeson SA (2015) Re-Os dating of pyrite confirms an early diagenetic onset and extended duration of mineralization in the Irish Zn-Pb ore field. *Geology* 43:143–146
- Höll R, Kling M, Schroll E (2007) Metallogenesis of germanium – a review. *Ore Geol Rev* 30:145–180
- Horral KB, Hagni RD, Kisvarsanyi G (1993) Mafic and ultramafic plutons associated with the New Madrid Rift Complex – A possible major source of the copper-cobalt-nickel mineralisation of Southeast Missouri. *Econ Geol* 88:328–343
- ILZSG (2017) *World Directory: lead and Zinc Mines 2016*. International Lead and Zinc Study Group, Lisbon
- Irish G (2019) Lisheen mine data release. Accessed on 05th of March 2023 at <https://www.gov.ie/en/publication/d187c-lisheen-mine-data-release/>
- Jochum KP, Weis U, Stoll B, Kuzmin D, Yang Q et al (2011) Determination of reference values for NIST SRM 610–617 glasses following ISO guidelines. *Geostand Geoanal Res* 35:397–429
- Johnston JD (1999) Regional fluid flow and the genesis of Irish carboniferous base metal deposits. *Min Deposita* 34:571–598
- Johnston JD, Coller D, Millar G, Critchley MF (1996) Basement structural controls on Carboniferous-hosted base metal mineral deposits in Ireland. In: Strogen P et al (eds) *Recent advances in Lower Carboniferous Geology*, vol 107. Geological Society Publication No, London Geological Society, London, pp 1–21
- Kinnaird JA, Ixer RA, Barreiro B, Nex PAM (2002) Contrasting sources for lead in Cu-polymetallic and Zn-Pb mineralization in Ireland: constraints from lead isotopes. *Min Deposita* 37:495–511
- Klünder Hansen M, Makovicky E, Karup-Møller S (2003) Exploratory studies on substitutions in tetrahedrite-tennantite solid solution. Part IV. Substitution of germanium and tin. *N Jb Min Abh* 179:43–71
- Koch HA, Chew DM, Hitzman MW, Slezak P, Dunlevy E et al (2022) Characterisation of Irish volcanic ash layers using combined in-situ U-Pb and Lu-Hf analyses of zircon and/or apatite and its implication for the growth rate of the Waulsortian reef. *Goldschmidt 2022 Conference*, Honolulu, Hawaii
- Kyne R, Torremans K, Güven J, Doyle R, Walsh J (2019) 3-D modelling of the Lisheen and Silvermines deposits, County Tipperary, Ireland: insights into structural controls on the formation of Irish Zn-Pb deposits. *Econ Geol* 114:93–116
- Lang J, Meere PA, Unitt RP, Johnson SC, Solferino G et al (2020) The vein hosted copper deposits of the allihies mining area, Southwest Ireland – a new structural and chronological evaluation. *J Geol Soc Lond* 177:671–685
- Leach DL, Viets JB, Foley-Ayuso N, Klein DP (1995) Mississippi Valley-type Pb-Zn deposits. In: du Bray EA (ed) *Preliminary*

- compilation of descriptive geoenvironmental mineral deposit models. USGS, Reston, Virginia, pp 234–243
- Leach DL, Sangster DF, Kelley KD, Large RR, Garven G et al (2005) Sediment-hosted lead-zinc deposits: A global perspective. In: Hedenquist JW. (eds) One Hundredth Anniversary Volume, Society of Economic Geologists, Littleton, Colorado, pp 561–607
- LeHuray AP, Caulfield JBD, Rye DM, Dixon PR (1987) Basement controls on sediment-hosted Zn-Pb deposits: a pb isotope study of Carboniferous mineralization in Central Ireland. *Econ Geol* 82:1695–1709
- Lombaard AF, Günzel A, Innes J, Krüger TL (1986) The Tsumeb lead-zinc-copper-silver deposit, South West Africa/Namibia. In: Annhaeusser CRM, S. (ed) Mineral deposits of Southern Africa. Geological Society of South Africa, Johannesburg, pp 1761–1782
- Luo K, Cugerone A, Zhou M-F, Zhou J-X, Sun G-T et al (2022) Germanium enrichment in sphalerite with acicular and euhedral textures: an example from the Zhulingou carbonate-hosted Zn(-Ge) deposit, South China. *Min Deposita* 57:1343–1365
- Lusk J, Calder BOE (2004) The composition of sphalerite and associated sulfides in reactions of the Cu-Fe-Zn-S, Fe-Zn-S and Cu-Fe-S systems at 1 bar and temperatures between 250 and 535°C. *Chem Geol* 203:319–345
- Morrissey CJ, Davis GR, Steed GM (1971) Mineralisation of the Lower Carboniferous of Central Ireland. *Trans Inst Min Metall* 80:B173–B185
- Osbahr I, Krause J, Bachmann K, Gutzmer J (2015) Efficient and accurate identification of platinum-group minerals by a combination of mineral liberation and electron probe microanalysis with a new approach to offline overlap correction of platinum-group element concentrations. *Microsc Microanal* 21:1080–1095
- Oszepealski S (1999) Origin of the Kupferschiefer polymetallic mineralization in Poland. *Min Deposita* 34:599–613
- Paradis S, Hannigan P, Dewing K (2007) Mississippi Valley-type lead-zinc deposits. In: Goodfellow WD (ed) Mineral deposits of Canada: a synthesis of Major Deposit-types, District Metallogeny, the evolution of geological provinces, and Exploration methods. Geological Association of Canada, St John's, Newfoundland, pp 185–203
- Poetz S, Liu Y, Magnall JM, Vieth-Hillebrand A, Yang S et al (2022) Signals of low grade organic matter alteration in the Upper Permian Kupferschiefer (Spremberg area, Eastern Germany) – A by-product of copper mineralisation? *Org Geochem* 169:104421
- Riegler T, McClenaghan SH (2017) Authigenic potassic silicates in the Rathdowney Trend, Southwest Ireland: new perspectives for ore genesis from petrography of gangue phases in Irish-type carbonate-hosted Zn-Pb deposits. *Ore Geol Rev* 88:140–155
- Romer DM (1986) A note on the Aherlow Cu-Ag deposit, County Limerick. In: Andrew CJ et al (eds) Geology and Genesis of Mineral deposits in Ireland. Irish Association for Economic Geology, Dublin, pp 509–511
- Scott SD, Barnes HL (1971) Sphalerite geothermometry and geobarometry. *Econ Geol* 66:653–669
- Scott SD, Kissin SA (1973) Sphalerite composition in the Zn-Fe-S system below 300°C. *Econ Geol* 68:475–479
- Shearley E, Redmond P, King M, Goodman R (1996) Geological controls on mineralisation and dolomitisation of the Lisheen Zn-Pb-Ag deposit, Co. Tipperary, Ireland. In: Strogon P (eds) Recent Advances in Lower Carboniferous Geology, Geological Society Special Publications 107, pp 23–33
- Silva AA, Cordeiro P, Johnson SC, Lagoeiro LE, Corcoran L et al (2022) The carbonate-hosted Tullacondra Cu-Ag Deposit, Mal-low Ireland. *Minerals* 11:560
- Steed GM (1986) The geology and genesis of the Gortdrum Cu-Ag-Hg orebody. In: Andrew CJ et al (eds) Geology and Genesis of Mineral deposits in Ireland. Irish Association for Economic Geology, Dublin, pp 481–499
- Strashimirov S, Dobrev S, Sivilova I (1999) Petrological and fluid inclusion studies of core samples from the Lisheen Mine, Ireland. Contract No. ML 582. Technical report. University of Mining and Geology St. Ivan Rilski, Sofia, Bulgaria, p 102
- Sun G, Zhou J-X, Cugerone A, Zhou M-F, Zhou L (2023) Germanium-rich nanoparticles in Cu-poor sphalerite: a new mechanism for Ge enrichment. *Geol Soc Am Bull* 136:2891–2905
- Swinkels L, Burisch M, Rossberg CM, Oelze M, Gutzmer J, Frenzel M (2021) Gold and silver deportment in sulfide ores – a case study of the Freiberg epithermal Ab-Pb-Zn district, Germany. *Min Eng* 174:107235
- Torremans K, Kyne R, Doyle R, Güven JF, Walsh JJ (2018) Controls on metal distribution at the Lisheen and Silvermines deposits: insights into fluid flow pathways in Irish-type Zn-Pb deposits. *Econ Geol* 113:1455–1477
- Van den Boogaart KG, Tolosana-Delgado R (2013) Analyzing compositional data with R. Springer, Berlin
- Warr LN (2021) IMA-CNMNC approved mineral symbols. *Mineral Mag* 85:291–320
- Watling RJ (1976) Trace element distribution in primary sulphide minerals from the Keel Prospect, County Longford. *Proc Roy Ir Acad* 76:241–261
- Wilkinson JJ (2010) A review of fluid inclusion constraints on mineralisation in the Irish orefield and implications for the genesis of sediment-hosted Zn-Pb deposits. *Econ Geol* 105:417–442
- Wilkinson JJ, Hitzman MW (2015) The Irish Zn-Pb Orefield: the view from 2014. In: Archibald SM, Piercey SJ (eds) Current perspectives on Zinc deposits. Irish Association for Economic Geology, Dublin, pp 59–72
- Wilkinson JJ, Eyre SL, Boyce AJ (2005a) Ore-forming processes in Irish-type carbonate-hosted Zn-Pb deposits: evidence from mineralogy, chemistry, and isotopic composition of sulfides at Lisheen Mine. *Econ Geol* 100:63–86
- Wilkinson JJ, Everett CE, Boyce AJ, Gleeson SA, Rye DM (2005b) Intracratonic seawater circulation and the genesis of seafloor zinc-lead mineralisation in the Irish Orefield. *Geology* 33:805–808
- Wilkinson JJ, Stoffell B, Wilkinson CC, Jeffries T, Appold MS (2009) Anomalous metal-rich fluids from hydrothermal ore deposits. *Science* 323:764–767
- Wilkinson JJ, Crowther HL, Coles BJ (2011) Chemical mass transfer during hydrothermal alteration of carbonates: controls of seafloor subsidence, sedimentation and Zn-Pb mineralisation in the Irish carboniferous. *Chem Geol* 289:55–75
- Wilkinson JJ (2023) Tracing metals from source to sink in Irish-type Zn-Pb(-Ba-Ag) deposits. In: Andrew CJ et al (eds) Irish-type deposits around the world. Irish Association for Economic Geology, Dublin, pp 147–168
- Wilson SA, Ridley WI, Koenig AE (2002) Development of sulfide calibration standards for the laser ablation inductively-coupled plasma mass spectrometry technique. *J Anal Atom Spectrom* 17:406–409
- Yesares L, Drummond DA, Hollis SP, Doran AL, Menuge JF et al (2019) Coupling mineralogy, textures, stable and radiogenic isotopes in identifying ore-forming processes in Irish-type carbonate-hosted Zn-Pb deposits. *Minerals* 9:335